MILITARY SPECIFICATION

DETECTING - RANGING SET AN/AAS-33A

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the criteria for fabrication and acceptance of production models of the Detecting-Ranging Set hereinafter referred to as the DRS.

1.2 System classification. The DRS consists of the following: (acronyms used herein are shown in parenthesis)

a.	Receiver Group (RG)	3347100-110,-111
b.	Turret, Stabilized Platform (TSP)	3347110-110,-111
c.	Amplifier, Electronic Control (ECA)	3347600-110,-111
d.	Receiver, Infrared (FLIR)	3347120-110
e.	Compressor, Reciprocating	3347200-110
f.	Indicator, Infrared	3347700-110
g.	Receiver-Transmitter, Laser (LRT)	3347130-110
h.	Generator-Processor (LTE)	3347400-110
i.	Assembly, Cable W1 (PFN Cable)	3530539
j.	Receiver-Laser (FAC RCVR)	3347140-110
k.	Processor, Signal (LRE)	3347500-110
l.	Power Supply (LVPS)	3347300-110
m.	Control, Detecting-Ranging Set	3347800-110
n.	Control, Temperature	3347900-110

1.2.1 Receiver group. The Receiver Group (RG) (Item a.) consists of the Turret, Stabilized Platform (TSP) (Item b.), with the Receiver, Infrared (Item d.), the Receiver-Transmitter, Laser (Item g.) and the Receiver, Laser (Item j.) mounted therein.

1.3 Associated equipment. The DRS operates with the associated equipment listed in 6.6 and as defined in DL 1372AS 100 (formerly the Interface Control Document (ICD) GAC 128ISV-001).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein. Later revisions, amendments, Qualified Products Tests, and superseding documents which apply to parts, materials and technical processes may be used when preferred by the contractor. When a later issue is used, all applicable requirements of the later issue shall be used.

SPECIFICATIONS

FEDERAL

QQ-C-320 - Chromium Plating (Electrodeposited).

MILITARY

MIL-P-514 - Plate, Identification, Instruction and Marking, Blank.

MIL-C-5015 - Connector, Electric, Circular Threaded AN Type, General Specification for.

MIL-B-5087 - Bonding, Electrical, and Lightning Protection, for Aerospace Systems.

MIL-W-5088 - Wiring, Aerospace Vehicle.

MIL-E-5400 - Electronic Equipment, Airborne, General Specification for.

MIL-T-5422 - Testing, Environmental, Aircraft Electronic Equipment.

MIL-H-5440 - Hydraulic Systems, Aircraft Types I and II, Design, Installation, Requirements for.

MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys.

MIL-H-5606 - Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance.

MIL-C-6021 - Castings, Classification and Inspection of.

MIL-E-6051 - Electromagnetic Compatibility Requirements, Systems.

MIL-H-6083 - Hydraulic Fluid, Petroleum Base, for Preservation and Operation.

MIL-I-6181 - Interference Control Requirements, Aircraft Equipment.

MIL-C-6781 - Control Panel; Aircraft Equipment, Rack or Console Mounted.

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MIL-T-6845 - Tubing, Steel, Corrosion Resistant (304), Aerospace Vehicle Hydraulic System, 1/8 Hard Condition.

MIL-W-6858 - Welding, Resistance, Spot and Seam.

MIL-P-7788 - Panel, Information, Integrally Illuminated.

MIL-M-7997 - Motors, Aircraft Hydraulic, Constant Displacement, General Specification for.

MIL-I-8500 - Interchangeability and Replaceability of Component Parts for Aircraft and Missiles.

MIL-W-8604 - Welding of Aluminum Alloys: Process for.

MIL-W-8611 - Welding, Metal Arc and Gas, Steels, and Corrosion and Heat Resistant Alloys; Process for.

MIL-H-8775 - Hydraulic System Components, Aircraft and Missiles, General Specification for.

MIL-S-8802 - Sealing Compound, Temperature-Resistant, Integral Fuel Tanks and Fuel Cell Cavities, High-Adhesion.

MIL-A-8860 - Airplane Strength and Rigidity, General Specification for.

MIL-A-8861 - Airplane Strength and Rigidity, Flight Loads.

MIL-A-8863 - Airplane Strength and Rigidity, Ground Loads for Navy Procured Airplanes.

MIL-A-8865 - Airplane Strength and Rigidity, Miscellaneous Loads.

MIL-A-8866 - Airplane Strength and Rigidity, Reliability Requirements, Repeated Loads and Fatigue.

MIL-A-8870 - Airplane Strength and Rigidity, Flutter and Divergence.

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of.

MIL-M-18012 - Markings for Aircrew Station Displays; Design and Configuration of.

MIL-F-18264 - Finishes; Organic, Weapons Systems, Application and Control of.

MIL-T-18303 - Test Procedures; Preproduction Acceptance and Life, for Aircraft Electronic Equipment, Format for.

MIL-N-18307 - Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment, Including Ground Support Equipment.

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MIL-S-19500 - Semiconductor Device, General Specification for.

MIL-P-23377 - Primer Coating, Epoxy-Polyamide, Chemical and Solvent Resistant.

MIL-C-26482 - Connectors (Electric, Circular Miniature, Quick Disconnect, Environment Resisting), Recepticles and Plugs, General Specification for.

MIL-M-38510 - Microcircuits, General Specification for.

MIL-I-46058 - Insulating Compound, Electrical (for Coating Printed Circuit Assemblies).

MIL-W-46132 - Welding, Fusion, Electron Beam, Process for.

MIL-C-81511 - Connector Electrical, Circular, High Density, Quick Disconnect, Environment Resisting, and Accessories; General Specification for.

MIL-C-81659 - Connectors, Electrical, Rectangular Crimp Contacts, General Specification for.

MIL-C-81773 - Coating, Polyurethane, Aliphatic, Weather Resistant.

MIL-C-81907 - Coating System, Polyurethane, Aliphatic, Weather Resistant, Process for Application of.

MIL-C-83286 - Coating, Urethane, Aliphatic Isocyanate, for Aerospace Applications.

MIL-C-83723 - Connector, Electric (Circular, Environment Resisting), Recepticles and Plugs, General Specification for.

NAVAL AIR SYSTEMS COMMAND

SD-24 (Volume I) - General Specification for Design and Construction of Aircraft Weapon Systems.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-108 - Definition of and Basic Requirements for Enclosure for Electric and Electronic Equipment.

MIL-STD-130 - Identification Marking of U.S. Military Property.

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

MIL-STD-454 - Standard General Requirements for Electronic Equipment.

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MIL-STD-470 - Maintainability Program Requirements (for Systems and Equipment).

MIL-STD-471 - Maintainability Demonstration.

MIL-STD-681 - Identification Coding and Application of Hookup and Lead Wire.

MIL-STD-704 - Aircraft, Electric Power, Characteristics.

MIL-STD-781 - Reliability Tests Exponential Distribution.

MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production.

MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of.

MIL-STD-810 - Environmental Test Methods.

MIL-STD-882 - System Safety Program Requirements.

MIL-STD-965 - Parts Control Program.

MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

MIL-STD-2074 - Failure Classification for Reliability Testing.

MS-17321 - Meter, Time Totalizing Miniature Digital 28 Volt DC.

MS-17322 - Meter, Time Totalizing, Miniature Digital 115 Volt 400 Cycles.

MS-33583 - Tubing End, Double Flare, Standard Dimensions for.

MS-33584 - Tubing End, Standard Dimensions for Flared.

MS-33649 - Boss, Fluid Connection - Internal Straight Thread.

HANDBOOKS

MILITARY

MIL-HDBK-5 - Metallic Materials and Elements Aerospace Vehicle Structures.

DRAWING

MILITARY

DL 1372AS 100 - A-6E - Interface Control Document (ICD) for Detecting - Ranging Set (Formerly GAC 128ISV-001).

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production testing. This specification makes provisions for production testing.

3.2 Parts and materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectronic items shall conform to requirements specified herein.
b. Other electronic parts requirements shall conform to MIL-E-5400, except as modified herein.
c. When previously produced models of the equipment did not use non-repairable subassemblies, the design shall not be changed to employ non-repairable subassemblies without the approval by the designated government representative.
d. Non-electronic parts and materials requirements shall conform to the requirements of MIL-E-5400 except that those parts, materials and processes that are part of the structural design of the DRS, shall conform to the requirements of SD-24, Volume I. Selection of materials, excluding processes, shall be made from MIL-HDBK-5. The use of 2014T6XX, 707576XX and 7079T6XX material is prohibited on all machine parts. Weld joints shall be in accordance with Section 3 and 4 of the following specifications:
    1. MIL-W-6858
    2. MIL-W-46132
    3. MIL-W-8604
    4. MIL-W-8611
e. Castings shall be in accordance with MIL-STD-454, Requirement 21, and shall be classified and inspected in accordance with MIL-C-6021, except castings used in Item g. of 1.2 shall not be subjected to dye penetrant inspection.
f. Part selection within a manufacturers part type to obtain system performance shall not be permitted.

3.2.1 Semiconductor devices. Semiconductor Devices shall be chosen and applied as outlined in MIL-E-5400 and MIL-S-19500. All semiconductor devices, with the exception of detectors, emitters, crystals or other devices unique to the FLIR or Laser, shall be silicon unless otherwise approved by the designated government representative.

3.2.2 Microelectronic modular assemblies. When used, microelectronic modular assemblies shall meet the requirements of MIL-M-38510, and MIL-STD-965. Conformal coatings, encapsulants, embedments of potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.2.3 Subassemblies, maintenance. The electronic portions of the equipment shall be divided into maintenance subassemblies as defined in 6.9.4. Maintenance subassemblies shall be considered repairable.

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3.2.4 Finishes. The entire exterior surface of the RG (Item a. of 1.2) less windows, or as a minimum, that portion extending into the airstream of the aircraft, shall be finished as follows:

a. MIL-P-23377 Epoxy Primer mixed and applied in accordance with MIL-F-18264 and MIL-C-81907.
b. Topcoat with MIL-C-81773, MIL-C-83286 or equivalent Aliphatic Polyurethane coating mixed and applied in accordance with MIL-F-18264 and MIL-C-81907. Topcoat color shall be gray Color No. 36375 in accordance with Table II of MIL-C-81773 or MIL-C-83286.
c. All exterior markings on the RG shall be applied in such a manner as to meet the legibility and permanency requirements of MIL-STD-130. The color shall be in contrast to (b) above. The markings shall be visible when the RG is in STOW. The markings shall include precautions for safe stow range (for laser safety) and for hands-off TSP windows.
d. That portion of the RG not extending into the airstream may be finished as specified in 3.2.4.a. as a minimum. In the event of a conflict between this paragraph and 3.3.2.7 herein, the latter shall prevail.
e. Touch-up finishing, if required, may be performed in accordance with this paragraph or any alternate procedure approved by an authorized government representative.

3.2.4.1 Subassembly finishes. Item f. and m. of 1.2 shall be finished in accordance with MIL-E-5400 and the color shall be lusterless black in accordance with Color No. 37038 of FED-STD-595. Items c., e., h., k., l., and n. of 1.2 shall be finished in accordance with MIL-E-5400 and the color shall be gray in accordance with FED-STD-595, Color No. 36231. Discoloration of the paint of Item e. of 1.2 (Compressor) shall be considered acceptable. Item b. inside, d., g., and j. of 1.2 shall be finished in accordance with MIL-E-5400 and shall be painted with a low reflectivity color.

3.2.5 Sealing. The DRS shall as a minimum, be sealed as follows:

a. Pressurized areas, skin joint, faying surfaces and access closures shall be sealed in accordance with MIL-S-8802.
b. Sealing of surfaces for aerodynamic smoothness shall be in accordance with SD-24 (Volume I).

3.3 Design and construction. The equipment shall conform with the applicable requirements of MIL-E-5400 for design, construction, finishes, and workmanship unless otherwise specified herein. Component reference designations shall be employed in the DRS in accordance with MIL-STD-454, requirement 67. Soldering shall be in accordance with MIL-STD-454, requirement 5, except type RA Flux may be used for the detector assembly and for wave soldering.

3.3.1 Total weight. The total weight of the DRS, excluding aircraft cables, shall not be greater than 490 pounds (222 kilograms).

3.3.2 Reliability. The contractor shall conduct an organized reliability program in accordance with MIL-STD-785.

3.3.2.1 Operational stability. The DRS shall satisfactorily operate continuously or intermittently for a period of at least 85 hours without the

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necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating life. The DRS shall have a total operating life of 5,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified in the maintenance plan by the contractor in accordance with 3.3.7.1 and shall not be used without approval by the designated government representative.

3.3.2.3 Reliability in mean-time-between-failure (MTBF). (see 4.3.2).

3.3.2.4 Time totalizing meter. The units listed in table I shall contain time totalizing meters, as indicated. The meter readout shall be visible from the exterior of the equipment while in the normal mounting position, except Items e. and j. Meter installation shall be designed so that failure of the meter or its installation shall in no case result in the inability of the equipment covered by this specification to meet its performance requirements. A pulse counter shall be incorporated in the LRT to record the cumulative number of laser emissions.

TABLE I. Time totalizing meters.

Item	Unit Nomenclature	Quantity	Type of Meter
a.	Turret Stabilized Platform (TSP)	1	MS-17322
b.	Amplifier, Electronic Control (ECA)	1	MS-17322
c.	Receiver, Infrared 1/	1	MS-17322
d.	Compressor, Reciprocating	1	MS-17322
e.	Indicator, Infrared	1	MS-17322
f.	Receiver-Transmitter Laser (LRT) 1/	1	MS-17322
g.	Processor, Signal (LRE)	1	MS-17321
h.	Generator-Processor (LTE)	1	MS-17322
i.	Power Supply (LVPS)	1	MS-17322
j.	Control, Detecting-Ranging Set	1	MS-17322

1/ Access to items c. and f. shall be available when lower half of gimbal ball is removed.

3.3.2.5 Temperature alarm. Thermally actuated safety devices shall be employed to provide a contact sense of overheating of the equipment listed in 1.2 with the exception of Items d., i., j., k., m., and n. Thermal actuation by failure of these safety devices shall not cause the equipment, with the exception of the Compressor, TSP heater blanket, LTE, and LRT, to fail to operate. Indication of an overheating condition shall be an output available at the equipment connectors. The temperature alarm signal interface is defined in DL 1372AS 100.

3.3.2.6 Grounding. The DRS signal ground configuration shall be as specified in DL 1372AS 100. The DRS prime power neutral/return shall be grounded to the airframe external to the DRS.

3.3.2.7 Bonding. The design requirements of MIL-B-5087, exclusive of Lightning Protection Test, shall apply for the DRS electrical bonding. Surfaces requiring electrical bonding shall be free of paint and the finish shall be

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chromate conversion coat in accordance with MIL-C-5541, Class B, or chromium plated in accordance with QQ-C-320, Class 2B, as applicable. When a conflict between MIL-B-5087 and 3.2.4 herein appears, the requirements of MIL-B-5087 shall prevail.

3.3.3 Cabling and connections.

3.3.3.1 Cables and connectors. The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400. Connectors used for Weapon Replaceable Assembly (WRA) interfaces shall be in accordance with MIL-C-81511, except for special connector requirements specified in DL 1372AS 100. In the event other connector insert arrangements are required, insert arrangements in accordance with the following specifications, and in the order of preference listed, may be used subject to prior approval by the designated government representative:

a. MIL-C-26482, Series 2 Crimp.
b. MIL-C-5015, MS-3450 Series.
c. MIL-C-83723 or MIL-C-26482, Series 1 and 3
d. Rectangular connections shall be in accordance with MIL-C-81659, Supplement 1.
e. When nonstandard contacts (i.e., shielded or coaxial) are used in otherwise standard MIL-C-81511, MIL-C-26482, or MIL-C-5015 connectors, the resultant connectors shall meet the Performance (Section 3) and Quality Assurance Provision (Section 4) of the specification required for the standard contact assembled connector.

3.3.3.2 Interconnection cabling. The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be in accordance with DL 1372AS 100.

3.3.3.3 Wire selection (hookup). The selection of hookup wire shall be in accordance with MIL-E-5400.

3.3.3.4 Wire coding (hookup). Wire coding for hookup wire shall be in accordance with MIL-E-5400, MIL-STD-454 (Requirement 20), and MIL-STD-681 (System 1 or System 3).

3.3.3.5 Spare terminals. All WRA connectors shall meet Requirement 10 of MIL-STD-454 for spare terminals.

3.3.3.6 Crimp contact connectors. When crimp contact environmentally sealed connectors are used, all contact positions shall be filled with contacts. Sealing plugs shall be inserted in the grommet holes of unused contacts.

3.3.4 Control panels. All rack or console mounted control panels shall conform to the applicable requirements of MIL-C-6781. The panel markings shall be in accordance in MIL-M-18012. Controls shall conform to the requirements of MIL-STD-1472.

3.3.4.1 Control panel illumination. The Control Panel shall have red integral lighting and the capability of being dimmed. The lighting shall conform to the requirements of MIL-P-7788 for Class IR panels. The intensity control for

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the panel lighting shall be accomplished in a separate console control not to be supplied as part of the equipment. Panel lighting shall use 5 VAC, 400 Hz aircraft power as specified in DL 1372AS 100.

3.3.4.2 Operation of controls. All controls shall operate smoothly without binding or sticking and without excessive backlash, freeplay or friction. Switches shall make positive non-intermittent contact at the indexed positions. Variable controls (e.g., potentiometers) shall perform their designated functions smoothly and continuously without producing any observable noise on the displays and without intermittent operation or evidence of stepped changes in the parameter varied.

3.3.5 Interchangeability. The equipment shall meet the interchangeability requirements of MIL-E-5400 except as modified herein.

3.3.5.1 System interchangeability. Equipment interchangeability shall exist between Items a. through n. listed in 1.2. The DRS shall operate without adjustment of controls not accessible to the operator in normal flight within the performance limits specified herein when any unit (Items a. through n. of 1.2) which has completed intermediate level maintenance as specified in 3.3.7.2.2 is interchanged. Items a., c., e., f., h., i., and k. through n. of 1.2, shall be capable of being interchanged at the organizational level of maintenance without degrading equipment performance and without the need for adjustment or use of standard or special test equipment.

3.3.5.2 Assembly/subassembly interchangeability. Interchangeability for each assembly and subassembly is defined in 6.9.2.

3.3.5.3 RG interchangeability. The RG (Item 1.2 a. herein) interchangeability shall be in accordance with MIL-E-5400 and MIL-I-8500.

3.3.6 Interference control. The generation of Electromagnetic Interference (EMI) by the equipment and the vulnerability of the equipment EMI shall be controlled within the limits specified in MIL-I-6181 and the requirements of this specification except for the Temperature Control Unit (Item 1.2.n. herein). (The design of the DRS shall preclude self-induced EMI and shall conform to MIL-I-6181 to the extent demonstrated during MIL-I-6181 EMI testing of 1975 and by MIL-E-6051 testing to the level of Navy conducted DT IIIA MIL-E-6051 tests of 1976).

3.3.6.1 Conducted and radiated emissions. The conducted and radiated emissions of the equipment shall be controlled within the limits and requirements specified in MIL-I-6181, as modified by paragraph 3.3.6.

3.3.6.2 Conducted and radiated susceptibility. The equipment shall not exhibit any undesired response when subjected to the tests and requirements specified in MIL-I-6181, as modified by paragraph 3.3.6. Any deviation from standard indication (equipment output with no susceptibility signal applied) shall be considered as an undesired response.

3.3.7 Provisions for maintainability. DRS maintainability shall be designed in accordance with the requirements stated herein, MIL-STD-470, and MIL-STD-1472.

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3.3.7.1 Scheduled maintenance. Scheduled maintenance shall be approved in the maintenance plan for equipment requiring servicing or replacement of parts in accordance with 3.3.2.2 shall either be required for safety of flight reasons, to enhance the operational capability of the aircraft, or be justified on the basis of cost effectiveness. The contractor shall specify in the maintenance plan the frequency (interval) of the scheduled maintenance requirement.

3.3.7.2 Unscheduled maintenance. Unscheduled maintenance shall be completed in accordance with the following:

3.3.7.2.1 Organizational level maintenance. Organizational ('O') level maintenance is that maintenance performed by the using activity on board the aircraft and consists of preparation for the repair action, fault detection, fault isolation to the defective WRA, WRA removal and replacement, verification and securing from the maintenance action. The summation of the times required for the maintenance action is the 'O' level repair time. To assist the maintenance at the 'O' level, BIT features are required to accomplish Operational Readiness Tests (ORT) for fault detection and Fault Isolation Tests (FIT) for fault isolation to the defective WRA. After accomplishment of the maintenance action, BIT shall be used to verify that the replacement of the defective WRA has restored proper operation of the system. Except for adjustments of controls on the cockpit control panel no adjustments, calibration nor electronic support equipment shall be required for fault detection or isolation at the 'O' level.

3.3.7.2.1.1 Time to repair. The characteristics of the equipment shall be such that it shall be possible, in the case of failure, to perform all corrective organizational maintenance actions, other than combat damage repair, within a mean-time-to-repair (MTTR) of 30 minutes (not to exceed a maximum of 45 minutes) for items c., e., f. and i. through n. of 1.2. The MTTR for item a. shall be 50 minutes (not to exceed 90 minutes) assuming free and clear access to all connectors, fittings, mounting bolts, etc. The 30 and 50 minute MTTR times include fault detection, fault isolation, WRA removal and replacement, and verification that the DRS has been restored to operable status. The time to gain access to the removable WRA through aircraft structure, for removal of the cable assembly, for ground support equipment (GSE) installation and removal, and the warm-up/cool-down time are not included in the MTTR time.

3.3.7.2.2 Intermediate level maintenance. Intermediate level maintenance shall consist of repairing the WRA, including checkout and fault isolation to a replaceable assembly. All repair, adjustment and calibration of WRAs and subassemblies shall be accomplished at the intermediate level of maintenance. TSP cover removal for access to internal Shop Replaceable Assemblies (SRAs) shall be less than 10 minutes maximum. Time to repair shall include preparation for the repair action, removal and replacement, adjustment, and securing from the maintenance action. The time for removal of the cable assembly is not included in the MTTR time. Intermediate level test equipment shall not be a limitation. The intermediate level of repair shall further divided between assembly (WRA) and subassembly repair as follows:

3.3.7.2.2.1 WRA repair. WRA repair shall restore the equipment to operable status, and consist of replacement of defective subassemblies or repair of defective wiring, external to a subassembly, or a combination of both. Sufficient test points shall be provided so that, in conjunction with normal WRA interface connections, it is possible to fault isolate to a subassembly without removing or

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disconnecting any subassembly or access covers with the exception of the TSP cover. All adjustments, monitoring and test points other than GSE connectors shall be placed on only one side of the exterior surfaces of the assembly, other than the mounting surface. Exceptions shall be approved by the designated government representative. The MTTR of the WRAs, excluding the RG, shall not exceed 50 minutes. The MTTR of the RG shall not exceed 200 minutes.

3.3.7.2.2.2 Subassembly repair. Subassembly repair shall restore the equipment to operable status and consist of replacement of defective non-repairable parts, repair of defective wiring on a subassembly or a combination of both. In so far as practical, all adjustments shall be made at the subassembly level. The MTTR of the subassemblies shall not exceed 70 minutes.

3.3.7.2.3 Depot level maintenance. All repair, adjustment and calibration of WRA or subassemblies not performed at intermediate level maintenance facilities shall be accomplished at the depot maintenance facilities. Items for depot maintenance shall be identified and the reason for the depot requirement technically justified in the maintenance plan.

3.3.7.3 Built-in-test (BIT). The BIT features of the DRS shall be designed to detect and isolate DRS failures at the '0' level of maintenance. These tests shall be capable of being performed from the aircraft cockpit by one operator of intermediate or lower skill. No written materials, other than a single detailed checklist, shall be required. BIT shall not affect mission performance (except while the tests are being performed) and shall be capable of being performed while airborne or on the ground. The Temperature Control (Item n. of 1.2) shall contain an independent, manually-operated BIT self-test switch and indicator light remotely located from the cockpit. No BIT is required for the PFN Cable (Item i. of 1.2).

3.3.7.4 BIT general requirements. The DRS BIT circuits shall meet the following detailed requirements:

a. All DRS BIT controls and readout devices shall be remotely located in accordance with DL 1372AS 100.
b. The DRS shall have provisions to prevent laser emission at unsafe levels from the RG during any BIT tests requiring laser firing. Positive interlocks shall be provided by the DRS, and checked via BIT, to verify that the laser is pointed in a "safe" position (in STOW) prior to tests involving laser firing (see 3.3.16.2). If the stow interlock is missing, the DRS shall:
   1. Inhibit laser firing,
   2. Light the NO-GO lamp on the aircraft BIT panel,
   3. Light the Interlock Failure lamp, and
   4. Stop the test.

3.3.7.5 Operational readiness test (ORT) requirements. The DRS ORT shall be capable of correctly determining the system operational status. ORT shall be capable of being run in 5 minutes after obtaining FLIR detector cooldown. Tests shall include the following:

a. DRS-0
b. DRS-1
c. DRS-2

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d. DRS-3
e. DRS-4
f. DRS-5
g. DRS-6

3.3.7.6 Indication of test results shall be provided by the DRS, the Aircraft BIT Panel, and in the case of DRS-5 and DRS-6, by the Aircraft Computer.

3.3.7.6.1 ORT DRS-0. DRS-0 shall test the BIT circuitry within the Laser Subsystem, the ECA (including stow logic and interlock fail logic), the FLIR Receiver (detector temperature sensing circuits in the SDC), and the Indicator (FLIR mode test pattern generated in the Indicator). The laser shall not be fired as part of this test.

3.3.7.6.2 ORT DRS-1. DRS-1 shall test the Laser Subsystem (by firng the Laser fifteen times), the ECA, and the Indicator (TV test pattern generated in the Indicator).

3.3.7.6.3 ORT DRS-2. DRS-2 shall provide a test of the entire Stabilization Subsystem and shall allow for a check of reticle control operation and symbology except for the FAC symbol.

3.3.7.6.4 ORT DRS-3. DRS-3 shall test the FAC Subsystem and the FLIR Thermal Computer.

3.3.7.6.5 ORT DRS-4. DRS-4 shall test the FLIR detector temperature "cool-down", FLIR normal scene video display and the ability of the TSP to respond to a boresight command.

3.3.7.6.6 ORT DRS-5. This test shall cause the TSP to respond to a sequence of AN/ASQ-155 slew commands. The DRS shall provide "DRS Test" and "Turret Tests" switch closures concurrently to indicate that this test has been selected. All laser firing shall be inhibited in this test. The FLIR and Indicator shall be on to provide the operator with normal FLIR video throughout this test.

3.3.7.6.7 ORT DRS-6. This test shall cause the Laser Subsystem to (1) fire a limited number of laser pulses when commanded by the AN/ASQ-155, (2) provide a binary range word (nominally 60,000 feet) to the AN/ASQ-155, and (3) determine correct reception of a dummy VPRF code word (defined in DL 1372AS 100) from the AN/ASQ-155.

3.3.7.7 Fault isolation test (FIT) requirements. The DRS shall have the capability to fault isolate detected failures to the correct WRA. Fault isolation shall employ BIT circuitry as well as a single check list. Isolation to a faulty WRA shall be accomplished within 5 minutes by positive indication without substitution or elimination of WRAs, with any exceptions approved by the designated government representative. Specific tests shall include the following:

a. DRS-7
b. DRS-8
c. FLIR-0
d. FLIR-1
e. FLIR-2
f. FLIR-3

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g. FLIR-4
h. TUR-0
i. TUR-1
j. TUR-2
k. LSR-0
l. LSR-1
m. LSR-2
n. LSR-6

3.3.7.7.1 FIT DRS-7. DRS-7 shall test power forms supplied to the DRS from the Low Voltage Power Supply (LVPS).

3.3.7.7.2 FIT DRS-8. DRS-8 shall verify Control Panel power switch functions and BIT logic.

3.3.7.7.3 FIT FLIR-0. FLIR-0 shall test the BIT circuitry within the FLIR Subsystem by verifying the temperature sensing circuits in the SDC and generating a FLIR mode test pattern on the Indicator.

3.3.7.7.4 FIT FLIR-1. FLIR-1 shall test the Indicator by generating a "TV" test pattern.

3.3.7.7.5 FIT FLIR-2. FLIR-2 shall verify FLIR detectors are "cooled down" and shall be independent of DRS operating mode selected.

3.3.7.7.6 FIT FLIR-3. FLIR-3 shall test the FLIR thermal compensator, ARC and for failed channels.

3.3.7.7.7 FIT FLIR-4. FLIR-4 shall test the FLIR Subsystem operation by using the visible scene at the Turret Boresight position.

3.3.7.7.8 FIT TUR-0. TUR-0 shall test if the TSP is in STOW.

3.3.7.7.9 FIT TUR-1. TUR-1 shall test the proper operation of the ECA.

3.3.7.7.10 FIT TUR-2. TUR-2 shall test the ability of the Stabilization Subsystem to accurately position the TSP.

3.3.7.7.11 FIT LSR-0. LSR-0 shall test the BIT circuitry within the Laser Subsystem.

3.3.7.7.12 FIT LSR-1. LSR-1 shall verify proper operation of the LTE by firing the laser.

3.3.7.7.13 FIT LSR-2. LSR-2 shall verify proper operation of the Laser Subsystem by firing the laser.

3.3.7.7.14 FIT LSR-6. LSR-6 shall verify performance of the FAC Subsystem.

3.3.7.8 Weapon replaceable assembly (WRA). As used herein, a WRA is an item which can be removed and replaced from the aircaft to repair a malfunction in the most expeditious and economical manner (see 6.9.3). Items a., c., e., f., h. and k. through m. of 1.2 are defined as WRAs for fault isolation purposes and shall be replaceable at the organizational level of maintenance.

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3.3.7.9 Materials and processes. Materials and processes for organizational or intermediate level maintenance shall not require uncommon skills, support equipment, techiques, or excessive maintenance task times.

3.3.7.10 Modularization. The WRAs shall be packaged in modular form easily removable for repair by replacement.

3.3.7.11 Support equipment compatibility. The WRA and subassembly shall contain the necessary test points and connectors for testing and isolating malfunctions.

3.3.7.12 Environment. The equipment design shall provide for organizational level maintenance in environments of any humidity, -54°C to +55°C, sand and dust, by personnel wearing clothing required of the particular environment. No degradation of performance shall be caused by required maintenance performed under the above conditions, nor shall unusual component or system cleanup be required because of the environmental conditions. The contractor shall include in the maintenance plan cleanup and special handling requirements concerned with maintenance in the range of environments stated.

3.3.7.13 Adjustments and alignments. The requirements for WRA adjustments or alignments at the intermediate level of maintenance shall be kept to a minimum. The WRAs shall be designed so that the adjustment/alignment points are accessible without disassembly of the WRA or subassembly with the exception of the indicator, which requires Printed Wiring Boards (PWB) to perform adjustments, and for removal of a TSP cover to gain access to the TSP inner gimbal adjustments. Adjustment/alignment shall not change with handling. All requirements for adjustments/alignments shall be part of the intermediate level WRA and subassembly test procedures.

3.3.7.14 Handles. Each Item of 1.2, except Item g. which shall have a grip finish, weighing more than 15 pounds (7 kilograms) shall have handles appropriately located to permit ease of removal and replacement from the aircraft, except for those units which have provisions for appropriate external handling fixtures. Handle designs shall be in accordance with MIL-STD-1472.

3.3.7.15 Conformal Coating. When conformal coating is required to meet the moisture requirements of environmental specifications, material in accordance with MIL-I-46058 shall be used, with the exception of Pockels cell fire circuit card assembly. In no case shall a heavy buildup of conformal coating be used to support parts in order to meet vibration requirements. Conformal coating shall be used on all printed circuit boards.

3.3.7.16 Accessibility. All assemblies and subassemblies forming a part of the equipment shall be designed to provide easy and ready access to internal parts, terminals, and wiring for complete circuit checking, and removal and replacement of component parts. Non-repairable units and assemblies designed specifically for fast removal shall be excluded from this requirement. It shall not be acceptable to displace or remove wires, cables, parts or subassemblies in order to gain access to test points, terminals, soldered connections, mounting screws, etc. When unavoidable, approval shall be obtained from the designated government representative.

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3.3.7.17 Reversibility. The equipments shall incorporate features such that it is mechanically impossible to install equipments which are not electrically compatible or to attach cables, tubes, electrical plugs or any other items in an improper manner. Mechanically keyed mating, different size connectors, etc., shall be used to eliminate all such possibilities. Shape of tubing, tie down provision, color codes, connector contacts, labelling, etc., shall not be used as primary methods of satisfying this requirement.

3.3.8 Nomenclature and nameplates. Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.3.8.1 Nameplate materials and marking methods. Nameplates shall be fabricated from reverse etched anodized aluminum or photo sensitive anodized aluminum. The quality of the finished nameplate shall be in accordance with MIL-P-514.

3.3.9 Standard conditions. The conditions stipulated in table II shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

TABLE II. Standard conditions.

Temperature:	Room ambient
Altitude:	Normal ground
Vibration:	None
Humidity:	Room ambient up to 90 percent relative humidity
Input Power Voltage:	115 VAC, 400 Hz shall meet the requirements of MIL-STD-704
Cooling Air:	Individual WRA flow shall be defined in DL 1372AS 100

3.3.10 Service conditions. The DRS shall operate within specified performance limits under any of the environmental service conditions or combination of conditions specified in MIL-E-5400 as shown in table III except as modified herein.

3.3.10.1 Vibration. The RG shall operate satisfactorily when subjected to the vibration input levels of figure 1 or 3 as applicable. All other items in 1.2 shall operate satisfactorily when subjected to the vibration input levels of figure 2.

3.3.10.1.1 Isolation. When it becomes necessary due to vibration, shock, or other considerations to include vibration isolators or resilient mounting devices in the design of the equipment, the lowest permitted resonant frequency (with equipment mounted on isolators) shall be 25 Hz. External isolators shall be integral. All vibration isolators and resilient mounts shall be subject to approval as specified in MIL-E-5400, prior to production tests.

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TABLE III. Service conditions.

Items of 1.2	Class
1.2 a., c., h., k., l.	2X (see 3.3.13)
1.2 e., i., n.	2
1.2 f., m. (cockpit mounted)	1A
1.2 d., g., j. (internal to item 1.2 b., forming item 1.2 a.)	

3.3.10.2 Temperature. The DRS shall operate satisfactorily when subjected to the following temperature conditions.

3.3.10.2.1 Temperature, aircraft compartments. All DRS WRAs shall operate satisfactorily when subjected to the temperature condition of MIL-E-5400 for Class 1A and Class 2 equipment, as depicted in table IV, except as specified herein. Cooling airflow shall be provided by the aircraft in accordance with 3.3.13.

TABLE IV. Temperature, aircraft compartments.

Event	Class 1A (Cockpit Mounted only)	Class 2 and 2X
Continuous operation:	-54°C to +55°C	-54°C to +71°C
Intermittent operation:	-54°C to +71°C	-54°C to +71°C
Non-operating:	-62°C to +85°C	-62°C to +95°C

3.3.10.2.2 Temperature, aircraft exterior. For flight operation only, the portion of the RG mounted external to the aircraft shall perform satisfactorily in the temperature/altitude environment defined by Curve 1 in figure 4 when supplied with conditioned air in accordance with 3.3.13.2 herein. Non-operating conditions and the ambient environment internal to the aircraft shall conform to 3.3.10.2.1 herein.

3.3.10.3 Altitude. Sea level to 40,000 feet operating and 50,000 feet non-operating for Class 2 and 2X equipments. Sea level to 30,000 feet for Class 1A equipments.

3.3.10.3.1 Pressurization. During operation, cockpit bleed air, as described in DL 1372AS 100, shall be supplied to the RG.

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3.3.10.4 Shock. The DRS shall be designed to withstand shock in accordance with the requirements of MIL-E-5400.

3.3.10.4.1 Design stiffness requirements. For components which are designed to be vibration isolated, the lowest natural resonant frequency shall be greater than 55 Hz to reduce structural magnification of the input through the isolation system. For components which are not vibration isolated, the lowest natural resonant frequency shall be greater than 45 Hz.

3.3.10.4.2 Constant acceleration. The equipment shall be capable of operating within specified performance limits when subjected to a continuous acceleration level of 6 gravity units (g) along the Z aircraft axis.

3.3.10.5 Electronic enclosures. The wheel-well mounted WRA (Item c. of 1.2) shall meet the splashproof requirements of MIL-STD-108.

3.3.11 Warm-up/cool-down. Power shall be applied to all WRAs during this period.

3.3.11.1 Warm-up time. The time required for the equipment, excluding the gyros and heater blanker, to warm up prior to operation shall not exceed 5 minutes under standard or extreme service conditions. The gyro warm-up time shall not exceed the limits of figure 5. The warm up time for heater blanket shall not exceed 180 minutes at -54°C, 30 minutes at -5°C and 17 minutes at +25°C.

3.3.11.2 Cool-down time. Without power applied to the FLIR for a period of 24 hours, the equipment shall meet operating requirements, within 17 minutes under standard conditions and 20 minutes under extreme conditions for nominal 115 V input power, and within 22 minutes under standard conditions and 25 minutes under extreme conditions for 108.5 V input power after power is applied to the detector cooling loop.

3.3.12 Primary input power requirements. The equipment shall meet all applicable requirements of MIL-E-5400 and shall give the specified performance when supplied with alternating current (AC) power having the characteristics of MIL-STD-704 for Category B equipment. The equipment shall operate satisfactorily within all limits of MIL-STD-704, except for transient conditions of figure 3 therein. Transient voltage conditions shall be as described in paragraph 3.3.12.1 herein. Maximum input power shall be 7.3 kVA, and power factor limits shall be in accordance with figure 12 of MIL-STD-704 for Category B equipment.

3.3.12.1 Undervoltage and overvoltage protection. The equipment shall not be damaged by input alternating current (AC) power voltages down to zero (0), nor up to 125 percent (156 volts root-mean-square (Vrms)) of the maximum individual phase voltage of table I of MIL-STD-704 for Category B equipment. The equipment shall automatically resume normal operation after any transient input voltage within the limits described above which shall not exceed those shown in figure 3 of MIL-STD-704 as follows:

a. Low Voltage (108 Vrms to 0 Vrms) - Curve 3
b. High Voltage (121 Vrms to 156 Vrms) - Curve 2
c. Limits of (1) and (4) of Figure 3 of MIL-STD-704 shall not apply

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3.3.12.2 Internal protection. The DRS shall contain the necessary internal safeguards so as to ensure that the failure of any individual WRA of the DRS will not result in damage to any other DRS or aircraft WRAs except for DRS control panel mode and BIT interface with other DRS units. Any failure in the subsystems other than AC or direct current (DC) power short circuits will not preclude operation of remaining non-failed subsystems.

3.3.13 Cooling. The design for cooling the equipment shall comply with the requirements of MIL-E-5400 as modified by DL 1372AS 100. The WRAs of the DRS, that require aircraft supplied forced air within the limits of 3.3.13.2 and 3.3.13.3 herein, shall have inlets and outlets in accordance with DL 1372AS 100.

3.3.13.1 Cooling method for equipment. Cold plate heat exchangers shall be utilized in conjunction with forced air cooling to meet the equipment cooling requirements. The cold plate shall be designed to preclude the leakage of cooling air containing moisture and contaminants into a WRA proper. In addition, the cold plate design shall meet the requirements of MIL-E-5400, and shall preclude the blockage of cooling airflow by accumulated moisture or ice.

3.3.13.2 Temperature of cooling air. The temperature of the cooling air provided by the aircraft is summarized in DL 1372AS 100.

3.3.13.2.1 Extreme temperature of cooling air. The DRS shall be capable of operating when the cooling air provided is in the range of -65 degrees Fahrenheit (^{o}F) (-54 degrees Centigrade (^{o}C)) to +120°F (49°C). Degraded performance is permitted when the air is outside the limits specified in 3.3.13.2 herein.

3.3.13.3 Maximum available flow rate. The maximum flow rates and associated pressure drop characteristics of cooling air supplied by the aircraft are defined in DL 1372AS 100.

3.3.13.4 Altitude pressure changes. The equipment shall be designed to sustain no structural or component physical or electrical degradation when subjected to positive and negative pressure changes. The surrounding air pressure may vary at a rate as high as 1.0 pound per square inch (psi) per second.

3.3.14 Design load requirements. The DRS shall withstand loads generated and imposed by the environmental and operational conditions specified in 3.3.10 and 3.3.13 as well as those specified in 3.3.14.2 herein except that cockpit mounted equipment (Items f. and m. of 1.2) need not comply with 3.3.14.2. The most critical combination of applicable loads shall be used in the design.

3.3.14.1 Strength requirements. Strength requirements for the design of the DRS except Items f. and m. of 1.2 herein shall be in accordance with MIL-A-8860, MIL-A-8861, MIL-A-8863, MIL-A-8865, MIL-A-8866 and MIL-A-8870.

3.3.14.2 Inertial and aerodynamic loads. The factors listed in table V shall be applied to obtain limit inertia loads for Conditions 1 through 4. The design maximum limit air load and pressure distribution are as shown in figure 6.

3.3.14.3 Installation preloads. In determining loads, consideration shall be given to the preloads imposed in mounting the DRS.

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TABLE V. Inertial and aerodynamic loads. (U)

No.	Condition 1/	N_x	N_y	N_z
1	Flight	±1.5	±2.5	-6.5
2	Flight	±1.5	±2.5	+4.7
3	Catapult	+5.0	±0.5	-0.67
4	Arrested Landing	-4.1	±2.5	-12.0

1/ Positive directions are up, aft, and port.

3.3.14.4 Pressurized structure. For the design of pressurized structures, the design pressures shall be the most critical combination of aerodynamic pressure and the pressure differential between the pressurized portions of the structure and the ambient atmosphere. The pressure differential to be used for each particular condition shall be either 1.33 times maximum attainable pressure differential or zero, whichever yields the more critical pressure for the particular condition.

3.3.14.5 Ultimate factor of safety. The minimum ultimate factor of safety to be used to arrive at ultimate loads shall be 1.5. Failure shall not occur when loads which are 1.5 times the limit are applied to all items except castings. Castings shall meet the requirements of 3.2.e. herein.

3.3.14.6 Yield factor of safety. The minimum yield factor of safety to arrive at yield loads shall be 1.15.

3.3.14.7 Allowable strengths. The calculated strength of single pieces of material or of an entire assembly shall be based on minimum values of mechanical properties listed in MIL-HDBK 5. Critical conditions shall be determined by the contractor from the requirements of 3.3.14 herein.

3.3.15 Hydraulic system requirements. The TSP outer azimuth gimbal drive shall be hydraulically driven. The hydraulic power drive system shall utilize hydraulic power provided by the aircraft combined hydraulic power system, and be capable of providing ample smooth power under constant, varying, or reverse loads without degraded or detrimental performance effects due to jitter, backlash, or instability over its design operating range. The azimuth drive unit shall be compatible with and not have its performance degraded when installed in the aircraft and operated within the specified environmental and operational conditions of 3.3.10, 3.3.13.2 and 3.3.14.2 herein.

3.3.15.1 Hydraulic system characteristics. The TSP hydraulic system shall meet the following requirements:

a. Fluid Type
MIL-H-5606
MIL-H-6083

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b. Inlet Pressure
Nominal - 2700 to 3000 psi
Maximum - 3560 psi (Full Flow System Relief)
The inlet pressure system or regulated inlet pressure system shall be designed to withstand the following pressures:
Proof - 4500 psi
Burst - 7500 psi

c. Inlet Fluid Temperature
Fluid - -65°F (-54°C) to +245°F (118°C)

d. Return Pressure
Nominal - 150 psi
The return system shall be designed to withstand the following pressures:
Proof - 3000 psi
Burst - 4500 psi

e. Servomotor Case Pressure
Nominal - 150 psi
Those portions of the hydraulic system contacting the servo motor case shall be designed to withstand the following loads:
Proof - 500 psi
Burst - 750 psi

f. Port Sizes

Inlet -	MS33649-4
Outlet -	MS33649-6
Case -	MS33649-6
Seal Drain -	MS33649-4

g. Flow Rate
3.1 gallons per minute (gpm) maximum
0.1 gpm minimum

h. Protective Screens
Nominal - 25 microns

i. Leak Rate
Seal Drain - 15 drops per hour maximum
RG Overboard Drain - 15 drops per hour maximum

3.3.15.2 System design. The hydraulic system shall meet the design requirements of MIL-H-5440 for a Type II 3000 psi class system except as modified herein.

3.3.15.2.1 RG overboard drain. The overboard drain system shall have an operating temperature and pressure equal to the ambient atmospheric temperature and pressure. The overboard drain system shall be separated from the case drain system by means of dynamic seals at all times. The case drain leak rate shall not exceed that specified in 3.3.15.1.i. herein. The aircraft exit of the overboard drain line shall be located such as to preclude the possibility of fluid draining onto the external portions of the TSP. Overboard seal leakage during normal operation or non-operation shall not exceed that specified in 3.3.15.1.i. herein.

3.3.15.2.2 Sealing. Sealing shall be provided such that any leakage from the motor shaft seal will be dumped overboard and at least two seals will exist between any fluid under pressure and the inner surfaces of the RG. Servomotor shaft sealing shall be such that any leakage will not come in contact with gear teeth or bearings. Any seal material contacting the hydraulic fluid shall not suffer deterioration when subjected to all hydraulic requirements herein.

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3.3.15.2.3 Thermal expansion. The TSP shall provide for the thermal expansion of the fluid when the system is disconnected. No thermal expansion device shall be required when the system is fully connected.

3.3.15.2.4 Quick disconnect interface. The interfaces between the aircraft combined system and RG outer azimuth gimbal drive shall be accomplished with the use of quick disconnects. The quick disconnects shall be self-sealing at both ends and shall not rely on system pressure for sealing. They shall be capable of sealing at 1 psi minimum as well as other specified system pressures. It shall be physically impossible to incorrectly couple the pressure, return, and case drain lines. Captive dust plugs shall be provided with the RG unit quick disconnects.

3.3.15.3 Servomotor design and performance requirements. The contractor shall meet the design and performance requirements of MIL-M-7997 except as modified herein.

3.3.15.3.1 Pressures. The servomotor shall meet the following requirements:

a. Pressure Port
   3000 - 2700 psi
b. Return Port
   Operating - 150 psi
   Proof - 3000 psi
c. Case
   Operating - 150 psi
   Proof - 500 psi
   Burst - 750 psi
d. Shaft Seal Leakage
   15 drops per hour maximum

3.3.15.3.2 Reverse transients. The servomotor shall be capable of reversing direction or stopping under maximum acceleration without damage and maintain peak transient pressures within 3000 psi.

3.3.15.3.3 Servicing. The servomotor shall be capable of being removed from the TSP for servicing without dismantling the TSP. It shall not be necessary to remove or displace additional components in order to gain access for servicing. Replacement shall not require alignments or adjustments to meet the interface requirements. The system shall be designed to prevent hydraulic fluid from entering or draining over the surface of the RG during servicing. Captive hardware shall be used to the maximum extent possible.

3.3.15.3.4 Protection. The servomotor shall be protected by a 25-micron protective screen.

3.3.15.3.5 Filtration. The RG hydraulic system shall be capable of normal operation when subjected to acceptable contamination levels as defined by MIL-H-5400. A 5-micron filter is provided in the aircraft hydraulic system as defined in DL 1372AS 100.

3.3.15.3.6 Servomotor case drain. A quick disconnect shall be provided for connecting to the aircraft supplied servomotor case drain line. The maximum operating pressure in the servomotor drain system will be less than 150 psi.

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3.3.15.4 Hydraulic components. Hydraulic system components shall be supplied in accordance with MIL-H-8775.

3.3.15.4.1 Servo control valve. A pressure control servo valve shall be provided.

3.3.15.4.2 Fluid connection bosses. Fluid connection bosses shall meet the requirements of MS33649.

3.3.15.4.3 Hydraulic tubing. Hydraulic tubing shall be corrosion resistant steel conforming to the requirements of MIL-T-6845. Tubing ends shall be flared in accordance with MS33583 or MS33584, as applicable.

3.3.16 Safety. Provisions shall be made in the DRS for safety of personnel and equipment during all phases of storage, transportation, handling, test operations, checkout, and maintenance. A system safety program in accordance with MIL-STD-882 shall be established to assure that safety is considered during the DRS production phase, taking into special consideration proposed design changes for possible safety degradation, and evaluation of reported failures during testing for safety impact.

3.3.16.1 Protection of ground personnel during organizational maintenance. The DRS shall include safety interlock provisions to prevent turret rotation during organizational maintenance, as specified in 3.3.7.2.1. To permit power on the turret for gimbal alignment during intermediate level maintenance, these interlock provisions shall be designed to permit temporary override during this activity.

3.3.16.2 Laser beam control. The contractor shall implement a laser safety program for manufacturing and testing of the DRS to prevent exposure to the DRS laser beam during operational use, manufacturing, and production testing. A "safe" (STOW) position for the laser is a position determined by the contractor such that laser firing during BIT shall produce no emission from the DRS exceeding 1.6 microjoules per square centimeter.

3.4 Performance. Deleted from production specification.

3.5 Detail requirements.

3.5.1 Stabilization subsystem. The Stabilization Subsystem, composed of the Turret, Stabilized Platform (TSP) and Electronic Control Amplifier (ECA), shall meet the following requirements:

3.5.1.1 Stabilization subsystem functions. The Stabilization Subsystem shall be capable of positioning and stabilizing the electro-optical sensor package (consisting of the FLIR, FAC RCVR and LRT) at a target on the ground in response to digital rate commands issued by the Aircraft Computer. The Stabilization Subsystem shall supply scaled gyro error signals to the FLIR Subsystem as Image Motion Compensation in order to adjust the Indicator display for low frequency inertial disturbances. The sensor package angular orientation shall be provided to the Aircraft Computer in synchro format and analog information to the FLIR Subsystem for Indicator display. Interfaces shall be in accordance with DL 1372AS 100 for Detecting-Ranging Set AN/AAS-33A.

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3.5.1.1.1 TSP. The TSP shall provide four gimbals for sensor package mounting, positioning and stabilization, cables for sensor operation and motors, amplifiers and feedback components for stabilization operation. The TSP shall provide environmental protection for the sensor package with an access for replacement of the following items:

a. FAC RCVR
b. LRT Flashlamp
c. Hydraulic Drive Motor
d. Motor Drive Amplifier (MDA)
e. Desiccants

3.5.1.1.2 ECA. The ECA shall provide the electronics for control of the Stabilization Subsystem. The servo response of the four TSP gimbals shall be controlled by the ECA based upon inertial reference values, Aircraft Computer commands or TSP position feedback in response to the selected operating mode. Additionally, the ECA shall provide electronic interface between the Aircraft Computer and Laser and FAC Subsystems for sequential digital information.

3.5.1.2 Operating modes. The Stabilization Subsystem shall operate in one of the following normal modes as determined by the DRS Control Panel switch positions:

a. OFF. With the TUR/FLIR power switch in the OFF position, the TSP shall rotate to the STOW position (defined in b. following) and all power to the TSP (except the aircraft supplied excitation to the synchros) and ECA shall be removed due to LVPS shutdown. An LVPS "ON" command shall be supplied by the ECA until the TSP reaches STOW or 30 seconds (nominal) has elapsed since selection of "OFF" on the DRS Control Panel.
b. STOW. With the TUR/FLIR power switch in the STOW position, the ECA shall be fully powered and, provided that the TSP is in the STOW position, only gyro heater and gyro spin motor power shall be applied to the TSP by the ECA. When in the STOW position, the gyros shall be caged. The STOW position shall afford protection to the windows for the sensors contained within the TSP. The STOW position shall be defined as that set of outer gimbal angles for which outer azimuth is 0 ± 1.5 degrees and outer elevation more negative than -200 degrees. A power off delay circuit in the ECA shall provide up to 30 seconds (nominal) for the TSP to reach the STOW position upon the selection of either the STOW or OFF positions of the TUR/FLIR power switch.
c. ON. With the TUR/FLIR power switch in the ON position, full power shall be applied to the Stabilization Subsystem and, dependent on the position selected on the TURRET MODE switch, one of the following three modes shall be commanded:
    1. Boresight mode. With the BRST position of the TURRET MODE switch selected, the TSP shall rotate to the nominal azimuth and elevation measured with respect to the TSP mounting surface.
    2. Computer and manual modes. With the TURRET MODE switch in the COMPT or MAN position, the TSP shall respond to rate input commands from the Aircraft Computer provided that it is on, signified by the Computer On Discrete signal in accordance with DL 1372AS 100. If this signal is absent, the TSP shall be commanded to the STOW position by the ECA.

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3.5.1.2.1 BIT. Upon receipt of a 28 VDC (nominal) test discrete, the Stabilization Subsystem shall go into BIT which meets the requirements of 3.3.7.3.

3.5.1.3 Contents. The Stabilization Subsystem shall contain the following assemblies and subassemblies:

a. Turret, Stabilized Platform (TSP)
b. Electronic Control Amplifier (ECA)

3.5.1.3.1 Turret, stabilized platform (TSP). The TSP shall consist of three major subassemblies: Inner Gimbal, Motor Drive Amplifier and Turret Assembly. The Inner Gimbal shall provide sensor mounting and primary gimbal axes movement. The Motor Drive Amplifier shall supply outer elevation, inner elevation and inner azimuth gimbal motor drive current in proportion to ECA drive command analog signals. The Turrent Assembly shall protect the sensor package from aerodynamic and environmental forces and shall provide the secondary set of outer azimuth and outer elevation gimbal axes. The Turret Assembly shall provide internal interconnecting cables for the servo system and external interface connections for electrical and cryogenic functions for the sensor package. The Receiver Group consists of the TSP, with the FLIR, LRT and FAC RCVR mounted therein.

3.5.1.3.1.1 Inner gimbal (IG). The Inner Gimbal assembly shall contain the following major subassemblies which supply motion and feedback for the servo subsystem:

a. Gyro Assembly
b. Synchro Assemblies (Az and El)
c. Torquer Assemblies (Az and El)
d. Shaft Az and El Housing Assembly
e. Inner Gimbal Cable
f. Inner Azimuth Potentiometer Assembly
g. El Pot Bracket Assembly

3.5.1.3.1.1.1 Gyro assembly. The Gyro assembly shall contain azimuth and elevation rate integrating gyros mounted such that their spin axes are parallel and input axes orthogonal to the ORL. The Gyro assembly shall be replaceable without the use of any adjustments or boresighting equipment. The gyro inertial response to TSP gimbal inertial motion in conjunction with a gyro torquing current proportional to computer rate command shall determine the rate error of TSP gimbal motion. The gyro signal output shall be proportional to the integrated rate error.

3.5.1.3.1.1.2 Synchro assembly. Two synchro assemblies shall determine the inner gimbal position for computer usage. One synchro mounted on the Inner Gimbal elevation axis shall define the elevation angle with respect to the TSP mounting surface and the other shall determine the inner azimuth axis.

3.5.1.3.1.1.3 Torquer assembly. Two torquer assemblies shall be used on the Inner Gimbal, one for inner elevation axis and the other for inner azimuth axis.

3.5.1.3.1.1.4 Shaft Az and El housing assembly. The Inner Gimbal shaft housing assembly shall locate the orthogonal azimuth and elevation axes and shall provide a mounting surface for the FLIR.

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3.5.1.3.1.1.5 Inner gimbal cable. The Inner Gimbal Cable shall provide interconnection between on-gimbal components and off-gimbal TSP wiring.

3.5.1.3.1.1.6 Inner azimuth potentiometer assembly. The Inner Az pot shall provide feedback to the ECA for the servo system, slaving of the outer azimuth gimbal and a part of the azimuth angle information for eventual display on the Indicator.

3.5.1.3.1.1.7 Elevation potentiometer bracket assembly. The inner elevation pot shall provide feedback to the ECA for the servo system, slaving of the outer elevation gimbal and the elevation angle information for eventual display on the Indicator.

3.5.1.3.1.2. Motor drive amplifier (MDA). The MDA shall consist of two subassemblies, the Outer Elevation Amplifier and El/Az Torquer Amplifier.

3.5.1.3.1.2.1 Outer elevation amplifier. The Outer Elevation Amplifier shall convert ECA analog outer elevation drive commands into proportional motor current for gimbal response.

3.5.1.3.1.2.2 El/Az torquer amplifier. The El/Az torquer amplifier shall convert ECA analog drive commands for both inner elevation and inner azimuth gimbal torquers into proportional current for gimbal response.

3.5.1.3.1.3 Turret assembly. The turret assembly shall provide outer azimuth and outer elevation gimbal axes, environmental and aerodynamic covering for the sensor package, interconnection cabling and cryogenic plumbing. Additionally, the TSP shall modify the sensor cooling air by means of a heat exchanger containing a heater blanket. The Turret Assembly shall contain the following subassemblies:

a. Outer Azimuth Assembly
b. Heat Exchanger Assembly
c. Hydraulic Motor Assembly
d. Desiccant Assembly

3.5.1.3.1.3.1 Outer azimuth assembly. The outer azimuth assembly shall provide the outer azimuth and outer elevation orthogonal axes of rotation including mechanical and electromechanical drive and feedback components. The mechanical drive components shall consist of bearings for rotation and gearing for motors, tachometers and potentiometers. The feedback components shall be outer elevation pot and tachometer, outer azimuth pot and tachometer for servo operation and an outer azimuth synchro for Aircraft Computer usage. Additionally, the Outer Azimuth Assembly shall contain the outer elevation motor, sensor and servo interconnection wiring, FLIR Compressor cryogenic interconnect plumbing, a fan to circulate TSP (sensor) cooling air and outer azimuth and inner gimbal zero pins/receptacles.

3.5.1.3.1.3.2 Heat exchanger assembly. The heat exchanger assembly shall provide a means of cooling or heating the air contained within the TSP. Heat transfer of the TSP air shall be accomplished by either heating Heat Exchanger fin stock or by allowing externally supplied air to extract heat from warm TSP air via the fin stock. Heating shall be accomplished by the Heater Blanket subassembly and controlled by the Temperature Control WRA.

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3.5.1.3.1.3.3 Hydraulic motor assembly. The hydraulic motor assembly shall utilize aircraft hydraulic power to move the outer azimuth gimbal in proportion to ECA drive current.

3.5.1.3.1.3.4 Desiccant assembly. A desiccant assembly shall be provided to reduce TSP cooling air humidity.

3.5.1.3.2 Electronic control amplifier (ECA). The ECA shall provide electronic processing of aircraft computer digital rate commands into TSP gimbal drive signals, reference and excitation voltages for TSP feedback components, NADIR logic, BIT processing, Mode Switching logic, Laser and FAC subsystems-to-computer interface, WRA test access, IMC voltage and azimuth and elevation gimbal angle reference for Indicator display. The major subassemblies of the ECA shall include the following:

a. Digital Interface Assembly
b. Azimuth and Elevation Rate Processing Assemblies
c. Azimuth Stabilization and Turret Electronic Assembly
d. Elevation Stabilization and Turret Electronics Assembly
e. Gyro Heater Controller Assembly
f. Control Logic Assembly
g. Gyro Driver and Oscillator Assembly
h. Relay Logic Assembly
i. ECA Chassis
j. Motor Protection Electronics

3.5.1.3.2.1 Digital interface assembly. This subassembly shall provide for utilization of Aircraft Computer serial data and clock rate for conversion to parallel digital data format required in the azimuth and elevation rate processing assemblies in accordance with DL 1372AS 100. Laser and FAC digital data to and from the Aircraft Computer shall be interfaced in this assembly in accordance with DL 1372AS 100.

3.5.1.3.2.2 Azimuth and elevation rate processing assemblies. Two of these subassemblies shall provide for conversion of parallel digital data to analog azimuth and elevation gyro torquer current output porportional to the rate commands from the Aircraft Computer. Additionally, the subassembly shall provide gyro caging circuits.

3.5.1.3.2.3 Az stabilization and turret electronics assembly. This subassembly shall process the azimuth gyro, tachometer and inner and outer position potentiometer outputs. Filtering, switching, amplification, buffering and servo compensation shall result in inner and outer azimuth drive signals, Az BIT voltages and NADIR control.

3.5.1.3.2.4 El stabilization and turret electronics assembly. This subassembly shall process the elevation gyro, tachometer and inner and outer potentiometer outputs. Filtering, switching, amplification, buffering and servo compensation shall result in inner and outer elevation drive signals, El BIT voltages and outer azimuth drive rate correction based on elevation angle.

3.5.1.3.2.5 Gyro heater controller assembly. This subassembly shall provide power control for the azimuth and elevation gyro heaters in the TSP gyro assembly. Heater power control shall be in response to requirements for heating detected by a sensor located in the rate gyros. The subassembly shall also

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provide amplification of the rate gyro output for use in Image Motion Compensation. In addition, the Gyro Heater Controller shall provide for stow, -70° nominal elevation angle and -90° elevation angle signals.

3.5.1.3.2.6 Control logic assembly. This subassembly shall provide BIT voltage comparison, BIT sequencing and pass/fail BIT logic for the ECA and TSP. Also, a turn-on delay to allow (TSP) gyro motor run-up shall be provided at initial DRS power application.

3.5.1.3.2.7 Gyro driver and oscillator assembly. This subassembly shall convert 28 VDC into 2-phase, 900 Hz for use as azimuth and elevation rate integrating gyro spin motor excitation and tachometer BIT reference. In addition, it shall convert 28 VDC into 4 kHz nominal for use as signal excitation voltage in the rate integrating gyros and as a phase reference for gyro error detection in the Stabilization and Turret Electronics Subassemblies.

3.5.1.3.2.8 Relay logic assembly. This subassembly shall provide relay logic, power distribution, a 30 second approximate delay timer to permit reaching STOW position after the TUR/FLIR switch is placed in the OFF position, ON-OFF command circuitry, +12 and -5 volt (DC) regulators, NADIR and Computer On Discrete logic signal processing circuits, mode control relays and the azimuth/elevation torquer drive disabling logic and relays.

3.5.1.3.2.9 ECA chassis. The ECA Chassis shall provide wiring interconnections, a thermal transfer cold wall, connectors mounting, input power switching relay and power amplifiers for the gyro heaters.

3.5.1.3.2.10 Motor Protection Electronics. This subassembly shall provide the sensing, timing and switching circuitry to limit the inner azimuth, inner elevation, and outer elevation motor drives when the drive signals exceed predetermined values for an excessive period of time.

3.5.1.4 Form factor. The overall dimensions and details of the TSP and ECA shall be in accordance with DL 1372AS 100.

3.5.1.5 Weight.

3.5.1.5.1 TSP. Weight of the TSP shall not be greater than 233 pounds (106 kilograms).

3.5.1.5.2 ECA. Weight of the ECA shall not be greater than 20 pounds (9 kilograms).

3.5.1.6 Mounting. Aircraft mounting and interface of the TSP and ECA shall be in accordance with DL 1372AS 100.

3.5.1.7 Electrical connections.

3.5.1.7.1 TSP. Electrical connections shall be provided to interconnect the TSP to other related and associated systems as specified herein. Output connections interfacing to aircraft wiring shall be as specified in DL 1372AS 100 and comply with 3.3.3.1.

3.5.1.7.2 ECA. Electrical connections necessary to interconnect the ECA to other related systems shall be as specified in DL 1372AS 100 and comply with 3.3.3.1.

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3.5.2 Receiver, Infrared (FLIR), Compressor, Reciprocating and Indicator, Infrared (FLIR). The FLIR Receiver, Compressor and FLIR Indicator (FLIR Subsystem) shall meet the following requirements:

3.5.2.1 Function. The FLIR Subsystem shall provide real time display of the 8-14 micron infrared scene imaged by its optical system. The FLIR sensor shall collect infrared radiation and focus it upon a linear detector array. A scanning mirror passes the IR scene over the detectors in a horizontal direction while an interlace mirror steps the image back and forth vertically. An operator controlled zoom allows the FOV imaged to be varied continously over a 5:1 nominal range. Individual detector outputs are AC coupled to individual amplifiers whose outputs are time partition multiplexed to form a continuous video stream displayable on a CRT with a high speed vertical raster.

3.5.2.1.1 Automatic control. Automatic gain control shall be employed to control the gains of individual detector-preamp channels.

3.5.2.1.2 Focus. FLIR focus is maintained during thermal excursions with a thermal computer that uses temperature inputs from the objective lens, lens #2, and the zoom structure to position lens #4 in the zoom to the proper location.

3.5.2.1.3 Video polarity. Provisions shall be incorporated to permit reversal of the video output polarity using the DRS Control Panel POLARITY switch. The WHITE hot mode shall permit targets hotter than the background to appear white on the display. The BLACK hot mode shall permit targets colder than the background to appear white on the display.

3.5.2.1.4 FLIR FOV symbol. A symbol shall be generated to provide the DRS operator with an indication of the FLIR FOV. The symbol shall appear both on the FLIR Indicator and in the FLIR video outputs. The symbol, as seen on the FLIR Indicator, shall be referenced to dull black FOV markers provided by a decal on the CRT face. (See figure 7.) The symbol as seen on the FLIR Indicator shall have the following characteristics:

a. The FOV shall be a horizontal marker displayed on the right side of the raster.
b. At FLIR minimum FOV (NFOV) the symbol shall be near the upper FOV reference marker. When the symbol is near the lowest FOV reference marker, it shall represent the FLIR maximum FOV (WFOV).
c. The symbol shall move in a vertical direction between the upper and lower reference markers. From top to bottom each subsequent reference marker represents an increase in FLIR FOV (referenced to NFOV) as shown in figure 7.

3.5.2.1.5 FLIR reticle. The real-time display and all video outputs shall include a reticle as shown in figure 8 to define the line-of-sight (LOS) of the DRS.

3.5.2.1.6 Gimbal angle indicators. The FLIR Subsystem shall accept TSP gimbal angular position information and generate two symbols on the FLIR Indicator and in the FLIR video outputs to indicate this information to the operator. The symbols, as seen on the FLIR Indicator, shall be referenced to dull black markers provided by decals on the CRT face. (See figure 7.) The intensity of the FOV marker reticle and symbols shall be continuously variable

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from full on to off and shall be controlled by the RETICLE control on the DRS Control Panel. The FOV marker, gimbal angle indicators (GAIs), reticle and FAC symbol shall always appear white, and shall be discernible on the FLIR Indicator with a non-saturated FLIR signal level regardless of FLIR polarity. These symbols are further described as follows:

a. Elevation symbol. The elevation symbol shall have the following characteristics when displayed on the FLIR Indicator.
   1. The symbol shall be a horizontal marker displayed on the left side of the raster.
   2. The symbol shall move linearly, in a vertical direction, as a function of the signal described in DL 1372AS 100. With 0 degree input, the symbol shall be positioned near the 0 degree reference marker. With a -180 degree input, the symbol shall be positioned near the -180 degree reference marker.

b. Azimuth symbol. The azimuth symbol shall have the following characteristics when displayed on the FLIR Indicator.
   1. The symbol shall be a vertical marker displayed along the bottom of the raster.
   2. The symbol shall move linearly, in a horizontal direction, as a function of the signal described in DL 1372AS 100. With 0 degree input, the symbol shall be positioned near the 0 degree reference marker. With a -180 degree input, the symbol shall be positioned left of the 0 degree reference marker. With a +180 degree input, the symbol shall be positioned right of the 0 degree reference marker.

3.5.2.1.7 FAC symbol. A FAC symbol shall be generated by the FLIR Subsystem to indicate that a laser-designated target within the FAC Subsystem FOV has been acquired. This symbol shall indicate target location relative to the DRS LOS using position information supplied by the FAC Subsystem. The symbol shall appear both on the FLIR Indicator and in the FLIR video outputs. The symbol as seen on the FLIR Indicator shall have the following characteristics:

a. The symbol shall be displayed only during the presence of the coded discrete signal described in DL 1372AS 100.
b. The symbol shall appear on the FLIR Indicator as a hollow square, with each side 0.5 inch (1.3 centimeters) long (nominal). The symbol size shall remain constant regardless of the FLIR zoom position.
c. The intensity of the symbol shall be continuously variable from ON to OFF. Intensity shall be controlled by the SYMBOL control on the DRS Control Panel. The symbol shall always appear white and shall be discernible on the FLIR Indicator against a non-saturated FLIR signal regardless of polarity.

3.5.2.1.8 Display interfaces. Video outputs shall be provided to the Analog Display Indicator (ADI) and Video Tape Recorder (VTR). The FLIR Indicator shall also be capable of displaying video from the VTR, CONDOR missile system, and the WALLEYE system. The FLIR Indicator shall contain six status/advisory lights to indicate system status as described in 3.5.2.3.3.12. Using signals that are supplied by the Stabilization Subsystem, the FLIR Indicator shall provide two-dimensional Image Motion Compensation (IMC) that is scaled as a function of the FLIR FOV. IMC shall be accomplished by moving the raster up and down and left to right in response to signals from the ECA indicating gimbal position. A Built-in-Test (BIT) capability of the FLIR Subsystem shall be provided. All signal interfaces and video formats shall be in accordance with DL 1372AS 100.

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3.5.2.2 Operating modes. The FLIR Subsystem shall have the following operating modes:

a. OFF mode
b. STOW (STBY) mode
c. ON mode
d. CONDOR mode
e. WALLEYE mode
f. VTR mode
g. BIT modes

3.5.2.2.1 OFF mode. In the OFF mode, no power shall be applied to the FLIR Subsystem.

3.5.2.2.2 STOW (STBY) mode. In the STBY mode, power shall be applied and the equipment operating normally except for the FLIR scanner and the Indicator sweep and video circuits.

3.5.2.2.3 ON mode. In the ON mode, all power shall be applied and the equipment operating as specified herein.

3.5.2.2.4 CONDOR mode. In the CONDOR mode, the Indicator shall display video information from the CONDOR missile system and provide this video information at the video output connection to the VTR. (See DL 1372AS 100 for video format and interface details).

3.5.2.2.5 WALLEYE mode. In the WALLEYE mode, the Indicator shall display video information from the WALLEYE weapon system and provide this video information at the video output connection to the VTR. (See DL 1372AS 100 for video format and interface details.)

3.5.2.2.6 VTR mode. In the VTR mode, the Indicator shall display encoded FLIR composite video, CONDOR video, or WALLEYE video as played back by the VTR. (See DL 1372AS 100 for video format and interface details.)

3.5.2.2.7 BIT modes. The FLIR Subsystem shall operate in 5 different BIT modes as specified in Paragraphs 3.3.7.7.3 through 3.3.7.7.7. These include a BIT self test of the FLIR Receiver (FLIR-0), two Indicator operational codes (FLIR-0, FLIR-1), a cryogenic performance check (FLIR-2), and two operational checks of the entire FLIR Subsystem (FLIR-3, FLIR-4).

3.5.2.3 Contents. The FLIR Subsystem shall contain the following assemblies and subassemblies:

a. Receiver, Infrared (FLIR)
    1. Sensor, Infrared
    2. Converter, Signal Data (SDC)
b. Compressor, Reciprocating
c. Indicator, Infrared

3.5.2.3.1 Receiver, infrared (FLIR). The Infrared Receiver shall consist of two major subassemblies: the Infrared Sensor and the Signal Data Converter. The Infrared Sensor shall contain the optical system, scanning assemblies and the Detector Assembly. The Signal Data Converter shall contain the electronics

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required to synchronize, amplify and multiplex to a single video line.

3.5.2.3.1.1 Sensor, infrared. The Infrared Sensor shall contain the major subassemblies listed below.

a. Housings and Optics
b. Zoom Assembly
c. Detector Assembly
d. Thermal Compensation Servo Electronics
e. Automatic Responsivity Control Narcissus Assembly (ARC/NARC)
f. Scan Mirror Assembly
g. Synchronizing Pulse Generator Assembly
h. Interlace Mirror Assembly
i. Zoom Counterweight Assembly

3.5.2.3.1.1.1 Housings and optics. The housings and optics shall contain the major parts and subassemblies listed:

a. Blower assembly - The blower assembly shall consist of a fan and connector used to convectively cool fin stock located at the base of the detector dewar assembly.
b. FLIR power harness assembly - The FLIR power harness assembly shall consist of the relay assembly, connectors and associated wiring. This assembly transmits power, BIT signals and FLIR control signals to the FLIR and transmits FLIR video, display synchronization signals and BIT results to the TSP.
c. FLIR receiver harness assembly (detector harness assembly) - The FLIR receiver harness assembly consists of flexible Conduit (Breeze Cables), connectors and associated assemblies, Detector Bias Circuit Board, Cross Coupling Circuit Board and wiring. The assembly consists of three cables with discrete connectors on one end and joined at the other into a U shaped junction box with two connectors which connect to the detector dewar assembly. Two cables (60 conductors each) conduct signals from the detectors to the SDC for amplification. The third contains bias wires (and return) carrying the bias signals to three zones of 59 detectors. The U shaped junction box also provides for passive cross-coupling of detector channels.
d. Objective lens assembly - The objective lens (Lens 1) assembly consists of an 8 inch nominal coated germanium lens which forms the FLIR entrance aperture and provides a seal to the FLIR casting. A thermistor located on the lens provides inputs to the thermal computer.
e. First folding mirror assembly - The folding mirror assembly consists of the First Folding Mirror and Machined Sensor Housing. The first folding mirror folds the optical path downward to the lower optics assembly.
f. Lower optics component assembly - The lower optics component assembly consists of the Lower Optics Assembly (Housing, Mirror No. 2 and Mirror No. 3), Lens II and III and Computer Electronics Thermal Compensation Assembly. IR energy focused into the lower optics assembly by the objective lens and first folding mirror is folded horizontally, reimaged and folded back up into the FLIR zoom assembly. Lens III forms a seal to prevent air from carrying dirt

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onto mirror 3. The other end of the sealed compartment mates to a bellows carried on the zoom structure. Breathing during zoom movement is facilitated through two filters located between lens III and mirror 3. The Computer Electronics Thermal Compensation Assembly receives inputs from thermistors located on lens I, lens II and the Zoom Assembly, manual focus inputs from the FOCUS control on the DRS Control Panel and Zoom position inputs to generate a command output for the Thermal Compensation Servo Electronics to position lens IV providing thermal compensation. Also included is BIT circuitry (activated by FLIR-3 or DRS-3) to check proper functioning of the Computer Electronics, Servo Electronics, Servo Motor, and Thermal Compensator Mechanism.

g. Connector mount assembly - The connector mount assembly consists of the Shield Assembly and Heater for ARC/NARC.

3.5.2.3.1.1.2 Zoom assembly. The zoom assembly provides a continously variable FLIR FOV. The zoom shall be controlled by a switch in the aircraft that interfaces with the DRS in accordance with DL 1372AS 100. The zoom assembly consists of a zoom structure and a zoom carriage mounted within it on three rotatable screws, and a bellows which seals the area between the lower optics assembly and lens elements #5 and #6. Zoom assembly movement shall be accomplished with a 115 V, 400 Hz, 3 phase motor that drives the screws. The zoom mechanism shall contain a potentiometer used to generate an analog signal indicating the FLIR FOV as defined in DL 1372AS 100 for use external to the DRS. In addition, a symbol shall be generated to provide an indication of the FLIR FOV on the Indicator in accordance with Paragraph 3.5.2.1. The zoom assembly shall contain three movable lens elements. Lens V and VI are mounted on the zoom carriage with a fixed relationship to each other. Lens IV is mounted to provide movement relative to Lens V and VI in addition to the movement provided by the zoom carriage. This differential movement is required to provide adequate focus in all FOVs over a static temperature range of 70°F (21°C) to 110°F (43°C). This movement is accomplished via a servo motor and cam on the thermal compensator assembly. The servo motor shall use the Thermal Compensation Servo Electronics output to maintain focus over the temperature range. A zoom thermistor shall provide an output to the Thermal Compensation Computer Electronics located in the Lower Optics Component Assembly.

3.5.2.3.1.1.3 Detector assembly. The detector assembly shall consist of a vacuum dewar containing the detectors and the separable split stirling drive cylinder/piston assembly used to cool the detector dewar.

3.5.2.3.1.1.3.1 Detector dewar. The detector dewar assembly shall be a permanently sealed vacuum dewar containing the detectors. The base of the dewar shall provide feed-throughs for detector leads and bias leads. The dewar shall provide thermal contact with the detector element. The dewar assembly shall form a complete detector assembly with the vacuum enclosure.

3.5.2.3.1.1.3.2 Cylinder/piston assembly. The cylinder/piston assembly shall consist of an expander cylinder with a removable gas spring. The cylinder shall house a two stage stirling expander piston using babbitt balls as the heat reservoir medium, and helium as the working gas. Access to the expander shall be by removal of the gas spring portion of the cylinder. The refrigerator shall be capable of maintaining the detectors at or below their operating temperature.

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3.5.2.3.1.1.4 Thermal compensation servo electronics. The Thermal Compensation Servo Electronics card shall contain the necessary circuitry to process the thermal compensation Command from the Computer Electronics located in the Lower Optics Component Assembly and develop the proper drive signal for Lens IV of the optical system. The signal shall position Lens IV at the point determined by the computer electronics to provide focus for the field of view selected and the static temperature conditions in the FLIR.

3.5.2.3.1.1.5 Automatic responsivity control narcissus assembly (ARC/NARC). The ARC/NARC assembly shall contain a Beam Splitter, Reflective Optical Chopper and Automatic Responsivity Control. This assembly and the heater of the Connector Mount Assembly shall provide the following functions.

a. Automatic responsivity control (ARC) - The ARC assembly injects a reference temperature chopped at 210 hertz (Hz) into the video by imaging the heated ARC source onto the FLIR detectors via a beam splitter. The 210 hertz signal is used to set the preamplifier gain to maintain a uniform sensitivity across the detector array.
b. Narcissus control (NARC) - The NARC source is matched to the inverse of the narcissus pattern generated by reflection of the detector by the optical elements back onto itself. When this NARC pattern is imaged onto the detector by the beam splitter, it reduces the narcissus effect.

3.5.2.3.1.1.6 Scan mirror assembly. The Scan Mirror Assembly shall contain the motor and mirror assembly to scan the image horizontally. The mirror, oscillating at 60 Hz shall be driven directly by a 360 Hz, 3 phase synchronous motor. The Scan Mirror Assembly shall also contain a Synchronizing Pick-Up assembly. The output of the Synchronizing Pick Up assembly is provided to the Synchronizing Pulse Generator, to provide horizontal and interlace synchronization for the FLIR video.

3.5.2.3.1.1.7 Synchronizing pulse generator circuit board assembly. The Synchronizing Pulse Generator card shall contain the necessary circuitry to process the Scan Mirror Assembly photo diode output and provide a pulse output for generating Horizontal Sync and Interlace signals in the SDC.

3.5.2.3.1.1.8 Interlace mirror assembly. The Interlace Mirror Assembly shall be a mirror motor assembly which provides the 60 Hz scan interlace, between successive FLIR fields.

3.5.2.3.1.1.9 Zoom counterweight assembly. The zoom counterweight assembly shall consist of a counterweight structure used to enclose the movable counterweight, the counterweight carriage and counterweight and the flexible shaft used to drive the counterweight using the zoom motor. The counterweight is located and driven such as to keep the FLIR center of gravity (CG) constant during FLIR zooming.

3.5.2.3.1.2 Converter, signal data (SDC). The Signal Data Converter shall contain the major subassemblies listed:

a. Chassis Assembly - Signal Data Converter
b. Preamp Multiplexer Circuit Board Assemblies (14 ea)
c. Built-In-Test Circuit Board Assembly

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d. Video Summer Circuit Board Assembly
e. Video Driver Circuit Board Assembly
f. Auto Responsivity Control Circuit Board Assembly
g. Vertical Logic Circuit Board Assembly
h. Horizontal Sync Circuit Board Assembly
i. FAC Output Circuit Board Assembly
j. Square and Marker Logic Circuit Board Assembly
k. A/D Encoder Circuit Board Assembly
l. Top Cover

3.5.2.3.1.2.1 Chassis assembly. The chassis assembly contains the SDC chassis used to mount all the SDC contents, the driver interconnection assembly used to transmit FLIR signals from the output side of the preamps, the preamp interconnection assembly used to carry signals from the signal input connectors to the preamps and the fan assembly used to cool the SDC.

3.5.2.3.1.2.2 Preamp multiplexer circuit board assemblies. The SDC shall contain fourteen Preamp Multiplexer Circuit Board Assemblies. Each card shall provide the circuitry necessary to amplify and time partition multiplex eight channels of video from the IR detector array.

3.5.2.3.1.2.3 Built-in-test circuit board assembly. The BIT Circuit Card shall contain the following circuitry:

a. Buffer Multiplexer Circuit which receives four time partition multiplexed video signals from the Preamp Multiplexer Cards. Each video input signal consists of the multiplexed outputs of the Preamp Multiplexer cards (28 channels). The output is a single buffered video line.
b. Built-In-Test circuits which include Detector Temperature and Thermal Computer BIT Logic and GO/NO GO indications.

3.5.2.3.1.2.4 Video summer circuit board assembly. The Video Summer Card shall contain the circuitry necessary to sum the symbology, FLIR reticle and ARC Cancellation inputs into the video. A video splitter circuit shall be provided for Black Hot video inversion.

3.5.2.3.1.2.5 Video driver circuit board assembly. The Video Driver Card shall contain the circuitry required to process the Video Summer and blanking signals into a final FLIR Video output signal for display on the IR Indicator. The card shall also contain a circuit to provide an output indication of SDC overtemperature.

3.5.2.3.1.2.6 Automatic responsivity control (ARC) circuit board assembly. The auto responsivity card shall contain the ARC driver card, used to control the ARC chopper amplitude and through it the gain of the system, and the ARC cancellation circuit used to cancel the ARC signal in the video by inserting an inverted signal at the ARC modulation frequency.

3.5.2.3.1.2.7 Vertical logic circuit board assembly. The Vertical Logic Card shall contain vertical reticle, horizontal reticle, vertical blanking and 6 MHz clock logic circuitry. Circuitry for FLIR Vertical Sync output to the IR Indicator shall also be contained in the Vertical Logic Card.

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3.5.2.3.1.2.8 Horizontal sync circuit board assembly. The Horizontal Sync Card shall provide the necessary circuitry for the following functions:

a. Horizontal Blanking
b. FLIR Horizontal Sync ouput for IR Indicator display
c. Interlace outputs for driving the Interlace mirror and for the IR Indicator display
d. Zoom Position Marker

3.5.2.3.1.2.9 FAC output circuit board assembly. The FAC Output Card contains the circuitry necessary for generatng the FAC Symbol when commanded by the FAC Subsystem.

3.5.2.3.1.2.10 Square and marker logic circuit board assembly. The Square and Marker Card shall contain the circuitry required for generation of the azimuth and elevation Gimbal Angle Indicators (GAIs). Also, circuitry to generate the vertical and horizontal triggers for the FAC Output Card shall be contained in the Square and Marker Card.

3.5.2.3.1.2.11 Analog-to-digital (A/D) encoder circuit board assembly. The A/D Encoder card shall contain A/D converter circuitry and circuitry for processing FAC and GAI elevation and azimuth position inputs.

3.5.2.3.2 Compressor, reciprocating. The Compressor shall provide helium pressure pulses to the Detector Assembly (see 3.5.2.3.1.1.3) to accomplish cryogenic cooling of the Detector array. The Compressor shall consist of dual opposing cylinders run from a central AC motor/crankshaft assembly. Cooling shall be provided by two fans located in shrouds covering the cylinder heads. Helium pressure, nominally 450 psi shall be monitorable using a helium gage attached at the helium charge port. Helium fill will be accomplished via a fill port in the area of the helium gage. The Compressor shall contain the following:

a. Machined Housing Assembly
b. Compressor Cylinder Assemblies
c. Duct/Blower Assemblies
d. Piston and Rider Assemblies (2 ea)
e. Tube Assembly
f. AC Motor and Capacitor Assembly
g. Compressor Shroud Assembly
h. Connecting Rod Assemblies (2 ea)
i. Helium Pressure Gage
j. Overtemperature Switch
k. Cover Assembly

3.5.2.3.2.1 Machined housing assembly. The housing shall be aluminum and shall contain the AC Motor, Crankshaft and Main Bearing under a nominal helium pressure of 450 psig.

3.5.2.3.2.2 Compressor cylinder assemblies. The Compressor shall contain two aluminum cylinders with steel sleeves. The pressure pulses produced in the cylinder shall be exhausted through six exhaust outlets located at the top of the cylinder and ducted to a single threaded outlet.

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3.5.2.3.2.3 Duct/blower assemblies. The Compressor shall be cooled by two fans which force air over each cylinder and the Compressor housing through a duct/shroud network.

3.5.2.3.2.4 Piston and rider assemblies. The Compressor shall contain two Piston and Rider Assemblies to produce pressure pulses in the corresponding cylinders. Each piston shall contain a two rider rings and four compression rings assembly.

3.5.2.3.2.5 Tube assembly. The tube assembly shall transmit the pressure pulses from each cylinder to a common manifold.

3.5.2.3.2.6 AC motor. A 3 phase, 115 Vrms, 400 Hz motor shall be required to drive the two opposing pistons via a crankshaft, connecting rods, and bearings. A set of three encapsulated capacitors shall be provided for power factor correction.

3.5.2.3.2.7 Compressor shroud assembly. A sheet metal shroud shall fit around cooling fins on the Compressor Housing and shall direct cooling air past the fins in order to keep the unit cooled.

3.5.2.3.2.8 Connecting rod assemblies. Two connecting rods shall be required to connect the piston to the crankshaft.

3.5.2.3.2.9 Helium pressure gage. A helium pressure gauge located at the fill port shall indicate the helium pressure in the compressor.

3.5.2.3.3 Indicator, infrared. The Indicator shall display the video information developed by the FLIR Receiver, the Video Tape Recorder (VTR), CONDOR and WALLEYE to the DRS operator and supply this video to the VTR and the Analog Display Indicator (ADI). In addition, the modes discussed in 3.5.2.2 shall be provided. The Indicator shall contain the following:

a. Electron Tube Assembly
b. High Voltage Power Supply
c. Deflection Module
d. Image Motion Compensator and Interlace Module
e. Sync Processor Module
f. Mode Logic and Sweep Fail Module
g. Slow Sweep and Video Module
h. VTR Encoder Module
i. VTR Decoder Module
j. Phase Lock Loop Module
k. Chassis
l. Front Panel Assembly
m. Fan

3.5.2.3.3.1 Electron tube assembly. This assembly shall consist of a magnetically shielded CRT with integrally mounted deflection yoke. The phosphor of the CRT shall approximate that of a standard registered P-4 type. An implosion mask shall be provided as an integral component of the Indicator to protect the operator from flying glass and debris due to implosion of the CRT.

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3.5.2.3.3.2 High voltage power supply (HVPS). This subassembly shall provide the CRT focus and grid supply voltage, the CRT anode voltage, and cutoff supply voltage.

3.5.2.3.3.3 Deflection module. This subassembly shall provide sweep power to the CRT deflection yoke, sweep interchange and adaptation from FLIR sweep format to television (CONDOR or WALLEYE) display requirements in accordance with DL 1372AS 100 descriptions. In addition, this subassembly provides an over-temperature signal and accepts both interlace and Image Motion Compensation (IMC) voltages, which are used to move the raster.

3.5.2.3.3.4 Image motion compensation (IMC) and interlace module. This subassembly shall provide Image Motion Compensation in the FLIR mode to the CRT Deflection Module as a linear response to vertical and horizontal TSP rate sensors as scaled by the FLIR FOV. Interlace control to the Deflection Module also shall be provided in FLIR, VTR, Condor and Walleye consistent with the deflection requirements of each sweep format as described in DL 1372AS 100.

3.5.2.3.3.5 Sync processor module. This subassembly shall accept video inputs in the TV mode and shall produce the following as output signals:

a. A "back porch" sync pulse to be used for gain control and DC restoration in video processing.
b. TV vertical sync pulse.
c. A "Phase Sample Gate" to establish the horizontal TV sweep rate and start point on the raster.

3.5.2.3.3.6 Mode logic and sweep fail module. This subassembly shall accept the inputs listed below, perform the required logic function and produce the outputs listed:

a. Inputs
   1. FLIR BIT Mode Selection
   2. TV BIT Mode Selection
   3. Condor Mode Selection
   4. Walleye Mode Selection
   5. VTR Mode Selection
   6. Decoded FLIR TV Mode Selection
   7. FLIR Vertical Sync
   8. TV Horizontal Drive
   9. Fast Retrace
   10. Slow Sweep
   11. Raster On
   12. Video Cutoff Storage
b. Outputs
   1. VTR Mode
   2. FLIR Sweep Encoder
   3. BIT TTL Encoder
   4. Condor Frequency Select
   5. VTR Frequency Select
   6. Interlace/IMC Enable
   7. Walleye TTL
   8. BIT Relay Drive
   9. VTR Relay Drive

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10. Condor TTL
11. Walleye TTL
12. FLIR Video Relay Driver
13. FLIR TTL
14. FLIR Vert Size Enable
15. FLIR Playback Vertical Size Enable
16. Walleye Horizontal Size Enable
17. TV Horizontal Size Enable
18. Yoke Relay Drive Enable
19. Interlace Enable
20. Sweep Fail Drive
21. Fast Sweep Drive

3.5.2.3.3.7 Slow sweep and video module. This subassembly shall provide switching of all video forms as controlled by operating mode logic. A video buffer shall provide an output of FLIR video forms to the ADI as described in the ICD. Raster brightness and TV contrast levels shall respond to external controls. A video amplifier/CRT driver shall be provided with a beam cut-off provision for sweep failure cut-off. DC restoration of all video signals except FLIR shall be implemented. The slow sweep drive signal shall be generated as a function of mode discrete logic inputs, slow sweep timing sync, feedback from the Deflection Module and FLIR horizontal IMC. This module shall also provide an output signal to the sweep failure protection circuit.

3.5.2.3.3.8 VTR encoder module. This subassembly shall provide encoded, composite video to the VTR and decoder, buffered ADI horizontal and vertical syncs, and selected syncs (TV or FLIR, decoded or actual), as a function of mode logic enable signals.

3.5.2.3.3.9 VTR decoder module. This subassembly shall decode the VTR playback signal to obtain video (FLIR or TV), interlace, and horizontal and vertical sync for display on the CRT.

3.5.2.3.3.10 Phase lock loop module. This subassembly shall provide (fast) TV drive for all TV modes in sync to the Phase Sample Gate (fast sync sweep). An adjustment for fast sweep sync control shall be provided. A FLIR mode signal path relay interupt of horizontal IMC shall also be provided. A "Raster ON" diode logic output shall result from input of "Indicator ON" "TV BIT" or "FLIR BIT" logic from the DRS Control.

3.5.2.3.3.11 Chassis. The chassis will house all previous subassemblies as well as connector assemblies, a card rack, covers, a fan, and wiring.

3.5.2.3.3.12 Front panel assembly. The front panel assembly shall contain a glass window, captive handle and six status/advisory lights to indicate system status. Each light shall be double lamp load, utilizing 28 VDC lamps, and shall be wired as shown in DL 1372AS 100. When not illuminated and viewed in normal ambient light, each light shall appear flat black in color, and the legends shall not be discernible. When illuminated, each legend shall appear aviation yellow. The lights shall be enabled as follows:

a. LASER light - This light shall illuminate for the following conditions.
   1. Steady illumination for greater than 65 ± 15% of rated laser output energy. This condition also applies during laser BIT.

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2. Blink at a rate of 2 Hz nominal for less than 65 ± 15% of rated laser output energy. During laser BIT, for this condition, the light shall be off.

b. L-PRF light - This light shall illuminate for the following conditions:

1. Steady illumination when the Laser Subsystem has not received any Variable Pulse Repetition Frequency (VPRF) words with valid parity from the computer. During laser BIT, except DRS-6, this condition indicates a successful internal check of the VPRF parity circuits. During DRS-6 this condition indicates parity check failure of a computer supplied special VPRF Code word.
2. Blink at a rate of 2 Hz nominal when the Laser Subsystem has received and stored a VPRF word with valid parity, but the most recent VPRF word has invalid parity. In all BIT tests, except DRS-6, this condition indicates malfunction of the VPRF parity check indication circuits. During DRS-6 this condition indicates either the special VPRF Code most recent update has bad parity or that there is a malfunction of the VPRF parity check indication circuits.
3. Off when the Laser Subsystem is OFF or when the Laser Subsystem is presently receiving VPRF words with valid parity. During laser BIT, except DRS-6, this condition indicates failure of the VPRF parity circuits check. During DRS-6 this condition indicates successful parity check of a computer supplied special VPRF Code word.

c. DES light - This light shall illuminate when the DESIGNATE mode is selected or when the laser switches to designate in the AUTO mode under computer control.

d. NO-GO light - This light shall illuminate for the following conditions:

1. Off when the FAC Subsystem is OFF or when the FAC Subsystem is presently receiving VPRF words with valid parity. Off during FAC BIT indicating failure of VPRF parity circuits check.
2. Blink at a rate of 2 Hz nominal when the FAC Subsystem has received and stored a VPRF word with valid parity, but the most recent VPRF word has invalid parity or when a VPRF word with valid parity has not been received in the last 400 msec (approximate). This condition does not apply during FAC BIT.
3. Steady illumination when the FAC Subsystem has not received any VPRF words with valid parity (this condition also applies during FAC BIT) or when the CONDOR Subsystem has a NO-GO condition or when the aircraft Analog-to-Digital (A/D) Converter has a NO-GO condition. Switching shall be accomplished external to the FLIR Indicator. Steady illumination during FAC BIT indicates successful VPRF parity circuits check.

e. RADAR light - This light shall illuminate when the computer is accepting radar sensor inputs and shall not illuminate when the computer is accepting TSP gimbal angle inputs.

f. RANGE light - This light shall be illuminated when:

1. CONDOR range is correct, or
2. Laser range information is being accepted by the Aircraft Computer. Switching shall be accomplished external to the FLIR Indicator. This condition also applies during BIT DRS-6.

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3.5.2.3.3.13 Fan. A fan shall cool the Indicator.

3.5.2.4 Form factor. The overall dimensions, locations of connectors, air inlets/outlets, ETIs, etc., of the FLIR Subsystem shall be in accordance with Infrared Receiver, Reciprocating Compressor, Infrared Indicator and DL 1372AS 100.

3.5.2.5 Weight. The weights of the Infrared Receiver, Reciprocating Compressor and Infrared Indicator shall not exceed 65, 34 and 25 pounds (30, 15 and 11 kilograms), respectively.

3.5.2.6 Mounting. The Infrared Receiver shall be mounted within the TSP. The Reciprocating Compressor and Infrared Indicator shall be mounted in accordance with the requirements of DL 1372AS 100.

3.5.2.7 Electrical connections. Electrical connections shall be provided as necessary to interconnect this subsystem to other related and associated subsystems as specified herein. The wiring within the Receiver Group shall be as specified by the Receiver Group schematic and wiring assembly drawings. WRA connections for the FLIR Subsystem shall be as specified by DL 1372AS 100 and conform to 3.3.3.1.

3.5.3 Receiver-Transmitter, Laser (LRT) , Generator-Processor (LTE), and Assembly, Cable (PFN Cable). The LRT, Generator-Processor and PFN Cable compose the Laser Subsystem and shall meet the following requirements:

3.5.3.1 Function. The Laser Subsystem shall be capable of transmitting laser pulses to determine range to a target or to designate a target with VPRF coded pulses. The Laser Subsystem shall have BIT and BIT self test capabilities.

3.5.3.2 Operating Modes. The Laser Subsystem shall have the following operating modes:

a. RECEIVE (REC) Mode
b. RANGE Mode
c. DESIGNATE (DES) Mode
d. AUTO Mode
e. BIT Modes

All mode selections except BIT shall be accomplished with the DRS Control as specified in paragraph 3.5.6. Provisions shall be made for an external interlock that shall break a required signal line necessary for laser firing. The interlock shall interrupt the magnetic hold-in circuit of the Laser Power Switch when in the ON position to automatically revert it to the REC position. LASER power switch will be bypassed in BIT mode. An interlock shall be provided to prevent high voltage generation if any of the laser assemblies are not electrically connected.

3.5.3.2.1 RECEIVE (REC) mode. In the REC Mode, the Laser Subsystem shall have power applied to all circuitry except for the high voltage (400 V and PFN) power supplies. The Laser Subsystem shall be capable of receiving Variable Pulse Repetition Frequency (VPRF) codes and determining if the received VPRF code has valid parity, but shall be inhibited from firing.

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3.5.3.2.2 RANGE mode. In the RANGE mode, the Laser Transmitter and Range Receiver shall be operational to determine range with each pulse initiated by an Iteration Start Pulse from the Aircraft Computer. The range computed by the Laser Subsystem shall be available in a register for interrogation by the Aircraft Computer until the next Iteration Start pulse. The Laser Subsystem shall inhibit laser firing in the RANGE mode if the Iteration Start pulse is not present.

3.5.3.2.3 DESIGNATE (DES) mode. In the DES mode, the Laser Transmitter shall be operational to provide a continuous VPRF coded train of laser pulses at any of the operator selected Tri-Service VPRF codes. Selection of the operating VPRF code shall be by means of the DRS/computer digital interface defined in DL 1372AS 100. If no valid VPRF code is received from the Aircraft Computer, the Laser Subsystem shall operate at a predetermined code and the L-PRF light shall illuminate steadily. If a code with invalid parity is received after reception of a valid code, the Laser Subsystem shall store, and operate at, the last valid code and the L-PRF light shall blink at a 2 Hz rate.

3.5.3.2.4 AUTO mode. The AUTO Mode shall be an Aircraft Computer controlled mode where the computer can select either RANGE mode or DES mode via PRF Switch Discrete described in DL 1372AS 100. When in AUTO mode, the Laser Subsystem shall operate as in the DES mode when the PRF Switch Discrete signal is applied. The absence of the PRF Switch Discrete signal shall cause the Laser Subsystem to operate in the RANGE mode when Iteration Start pulses are received.

3.5.3.2.5 BIT modes. The Laser Subsystem shall have a BIT self test and three laser firing BIT codes as specified in 3.3.7.6.7, 3.3.7.7.11, 3.3.7.7.12, 3.3.7.7.13.

3.5.3.3 Contents. The Laser Subsystem shall consist of the following:

a. Receiver-Transmitter, Laser (LRT)
b. Generator-Processor (LTE)
c. Assembly, Cable (PFN Cable)

3.5.3.3.1 Receiver-transmitter, laser (LRT). The LRT shall contain the collimating optics, resonator and beam steering optics, photodetector, and assemblies as follows:

a. Cooling Module
b. Forward Optics
c. Pockels Cell Assembly
d. Truncated Porro Prism
e. Transmitter Telescope
f. Receiver Telescope
g. Pockels Cell Electronics
h. Transmitter Logic
i. Receiver
j. Housing

3.5.3.3.1.1 Cooling module. The Cooling Module shall contain the laser rod, flashlamp and flashlamp trigger transformer, and reflective pump cavity. Cooling shall be accomplished by circulating a high pressure inert gas mixture past the flashlamp and laser rod and through the internal portion of a heat

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exchanger. Heat shall be extracted by a fan blowing outside air through the external portion of the heat exchanger. A pressure switch shall be incorporated in the high pressure area to provide a PRESSURE LOW signal, so that the LTE can inhibit laser firing in the event of loss of pressure. Temperature sensing shall be incorporated to generate an overtemperature signal so that the LTE can inhibit firing in the event of high overall temperature or excessive internal temperature compared to outside air temperature. A pressure gage shall be incorporated so that actual pressure value can be visually observed.

3.5.3.3.1.2 Forward optics assembly. The Forward Optics Assembly shall include mounting provisions and adjustments, where required, for the following: dielectric output reflector and resonator porro prism, a resonator alignment wedge, a transfer prism and dual right angle prism for directing the laser beam to the transmitter telescope, a beam splitter and diffuser to direct a small portion of the laser beam to the energy sense photodiode on the Transmitter Logic Assembly and a pair of individually adjustable alignment wedges for transmitter boresight adjustment. The attachment of optics to mounts shall be accomplished in a manner providing thermally stable optical alignment.

3.5.3.3.1.3 Pockels cell assembly. The Pockels Cell Assembly shall consist of a Lithium Niobate crystal, electrodes and polarizers in a ceramic housing. The assembly shall have provisions for adjustable mounting in the Optical Bench and mating with the Pockels Cell Electronics.

3.5.3.3.1.4 Truncated porro prism assembly. The Truncated Porro Prism Assembly shall consist of a mount and a prism used to fold the resonator beam 180 degrees. The assembly shall be adjustable and mounted to the Optical Bench.

3.5.3.3.1.5 Transmitter telescope. The Transmitter Telescope shall be a Galilean type beam expander to collimate the laser beam to meet the beam divergence requirements. A pumplight filter shall be included to preclude the emission of visible light.

3.5.3.3.1.6 Receiver telescope. The Receiver Telescope shall collect and collimate reflected laser light onto the photodetector of the Receiver Electronics Assembly. The telescope shall include adjustment capability to allow boresighting to the transmitter beam. The size of the field stop shall limit the field of view. A spectral bandpass filter shall be incorporated. A light pipe shall be attached to the Receiver Telescope to collect a portion of the outgoing transmitter beam.

3.5.3.3.1.7 Pockels cell electronics. The Pockels Cell Electronics shall contain circuits to generate the required Pockels Cell bias voltage pulse whenever triggered by an external pulse. The pulse width shall be controlled to preclude post-lasing. The high voltage bias pulse shall be generated from the +400 VDC input voltage available to the Pockels Cell Electronics.

3.5.3.3.1.8 Transmitter logic. The Transmitter Logic shall contain the following:

a. Circuitry to manually adjust a DC output voltage which is used to complement the voltage level of the energy control loop, by setting it in the center of its dynamic range for nominal laser conditions.

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b. A transmitter energy monitoring detector and signal processor to generate an error voltage proportional to the deviation from the specified laser output energy level and to provide an output pulse every time the laser fires to enable the Range Counter, and another output pulse each time the laser fires, provided the energy level is above its specified minimum.
c. LRT overtemperature monitoring and output logic signal circuitry.
d. Capacitor discharge circuit to provide a drive pulse to the flash lamp trigger transformer each time the flashlamp is to be triggered.
e. Built-in-Test circuits.
f. LRT interlock control circuitry.

3.5.3.3.1.9 Receiver. The Receiver shall contain the following:

a. Avalanche photodiode detector and detector bias control circuitry.
b. Detector signal preamplifier.
c. Preamplifier output to video logic, interface circuitry.
d. Receiver time programmed gain control circuit.
e. Reset timing logic for the Laser Receiver.

3.5.3.3.1.10 LRT housing. The housing shall include provisions for mounting all of the above mentioned components in addition to the covers, flashlamp shot counter, elapsed time meter, optical filters for the transmitter logic photodetector, an external mirror, and a pressure indicator.

3.5.3.3.2 Generator-processor, (LTE). The LTE shall consist of a chassis providing for heat exchange and containing the electronics assemblies listed below:

a. Logic Assembly
b. PFN Power Supply Assembly
c. PFN Assembly
d. Filter Relay Assembly
e. Range Counter Assembly

3.5.3.3.2.1 Logic assembly. The Logic Assembly shall contain all the logic and control circuitry to operate the Laser Subsystem in the proper mode and at the proper VPRF code according to Aircraft Computer and DRS Control inputs. It shall inhibit laser firing during conditions of overtemperature, interlock failure, or low pressure in the laser cavity. BIT control and self test circuitry for the Laser Subsystem shall also be included. The Logic Assembly provides the following outputs:

a. Blanking logic for use by aircraft mounted weapons.
b. LTE and LRT temperature sensing logic.
c. Laser status logic to drive DRS indicator lights.
d. BIT result logic to indicate laser GO/NO GO conditions to the DRS Control Panel.
e. Laser prefire logic to blank FAC RCVR input when lasing in the DES mode.
f. VPRF code interface logic for FAC RCVR clock, data and sync inputs.
g. Control logic for PFN and 400 VDC generation.
h. Count pulse per laser firing to shot counter.

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3.5.3.3.2.2 PFN power supply assembly. The PFN Power Supply Assembly shall contain the circuitry to generate the voltage to charge the PFN capacitor, utilizing the 3 phase 115 V 400 Hz power form. A temperature sensing device shall provide an output inhibit signal for LTE overtemperature conditions. The Power Supply Drive Module shall sum external adjustment of the PFN voltage, energy error voltage level, and PFN voltage level to allow dynamic control of the PFN voltage. Circuitry shall be included to shut down the PFN Power Supply Drive when commanded to do so by external input signals. Interlock logic and control circuitry shall be provided to discharge the PFN voltage when a Laser Subsystem interlock fail condition or loss of power to LTE occurs. Circuits shall be provided to generate a +400 VDC output from the +28 VDC input when commanded by an external input signal. BIT circuitry shall be included to provide an output to indicate when the +400 VDC and PFN voltage are below their minimum specification.

3.5.3.3.2.3 PFN assembly. The PFN Assembly shall contain the capacitance capable of storing the high energy required by the flashlamp and the inductance suitable for controlling the capacitor discharge characteristics into the flashlamp.

3.5.3.3.2.4 Filter relay assembly. The Filter Relay Assembly shall contain EMI suppression filters for all power forms used in the Laser Subsystem. In addition, it shall contain power control relays for energizing the Laser Subsystem.

3.5.3.3.2.5 Range counter assembly. The Range Counter Assembly shall contain the counting and resetting logic and clock required to measure time interval between initiation of the laser pulse and return of the last reflection of that laser pulse occurring within the maximum range specification. The range counter will process the time interval into a binary range word, store it, and make it available on command in serial format to the Aircraft Computer.

3.5.3.3.3 Assembly, cable (PFN cable). The PFN Cable Assembly contains no subassemblies. It is used to transfer the energy stored in the PFN Assembly in the LTE to the flashlamp in the LRT and to provide an interconnect sensing circuit.

3.5.3.4 Form factor. The overall dimensions, locations of connectors, air inlets/outlets, ETIs, etc., of the Laser Subsystem shall be in accordance with DL 1372AS 100.

3.5.3.5 Weight. The weights of the LRT, LTE and PFN Cable shall not be greater than 22, 26 and 1 pounds (10, 12 and 0.5 kilograms) respectively.

3.5.3.6 Mounting. The LRT shall be mounted within the TSP on the FLIR RCVR optical housing. The mounting provisions shall permit the interchangeability of units meeting the boresight accuracy requirements without requiring the use of any boresighting equipment. The LTE shall be mounted in accordance with the requirements of DL 1372AS 100.

3.5.3.7 Electrical connections. Electrical connections shall be provided as necessary to interconnect this system to other related and associated systems and conform to 3.3.3.1.

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3.5.4 Receiver, laser (FAC RCVR), and processor, signal (LRE). The FAC RCVR and LRE compose the FAC Subsystem and shall meet the following requirements:

3.5.4.1 Function. The FAC Subsystem shall be capable of detecting and indicating the angular position relative to the FLIR Optical Reference Line (ORL) of targets designated by VPRF coded lasers. The detected targets angular position shall be identified on the FLIR Indicator video scene by the FAC symbol, a white hollow, 0.5 in (1.3 cm) square. Whenever the detected target is within the FAC FOV, but outside the FLIR FOV, the FAC symbol shall limit near the edge of the FLIR display in the direction of the target. Detection of targets within the field of view and above the minimum detectable signal levels shall be dependent upon laser designation and a valid Variable Pulse Repetition Frequency (VPRF) Code. The FAC Subsystem, as modified by turret window vignetting shall have a definitive field of view (FOV) in both azimuth and elevation. The signal interfaces shall be in accordance with DL 1372AS 100.

3.5.4.2.1 Operating modes. The FAC Subsystem shall respond to the Control Laser Power Switch (3.5.6.2.1.3) as follows:

a. In the ON position, the FAC Subsystem shall be fully operational.
b. In the REC position, the FAC Subsystem shall be fully operational.
c. In the OFF position, the FAC Subsystem shall not be powered.

3.5.4.3 Contents. The FAC Subsystem shall consist of the following:

a. Laser Receiver (FAC RCVR)
b. Signal Processor (LRE)

3.5.4.3.1 Laser receiver (FAC RCVR). The FAC RCVR shall consist of a narrow bandpass spectral filter and an f/1.03, 2 element telescope assembly boresighted to and mounted alongside the FLIR sensor assembly, and a chassis providing a heat sink, interconnection wiring and containing the following electronic assemblies:

a. Detector and Preamplifier Assembly
b. Pulse Amplifiers
c. Peak Detector and Dump Pulse Generator
d. Filter and Gain Switch Assembly

3.5.4.3.1.1 Detector and preamplifier assembly. This assembly shall provide:

a. A quadrant silicon photodiode detector to receive pulsed laser energy and provide four output currents with relative magnitudes containing position information of the received laser energy with respect to the FAC RCVR ORL.
b. Transimpedance preamplifiers for each detector channel.
c. Gain switching circuits for each preamplifier channel to provide for dynamic range requirements.
d. A light emitting diode (LED) as an optical test source for quadrant detector illumination during BIT.

3.5.4.3.1.2 Pulse amplifiers. These two (2) assemblies contain four pulse amplifiers and shall provide for each preamplifier channel:

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a. Switchable attenuator for dynamic range requirements.
b. A pulse amplifier whose gain is controlled by AGC action.
c. DC restoration circuit.
d. Pulse delay circuit.
e. AGC pulse amplifier loop integration.

3.5.4.3.1.3 Peak detector and dump pulse generator. This assembly shall provide for each of the four detectors:

a. An amplitude peak detector and hold amplifier with reset capability and last pulse logic.
b. A low pass filter and output buffer for sample and hold amplifier signals.
c. An automatic gain control (AGC).
d. Threshold circuit operating on the summation of the pulse amplifier outputs.
e. A threshold blanking circuit.

3.5.4.3.1.4 Filter and gain switch assembly. This assembly shall provide a gain switching circuit for the preamps and, in addition, shall provide a BIT LED drive circuit.

3.5.4.3.2 Signal processor (LRE). The LRE shall consist of a chassis providing for heat exchange, interconnection wiring, power relays, and containing the following electrical assemblies:

a. Video Processor Assembly
b. Timing Logic Assembly
c. BIN Logic Assembly
d. Track Logic Assembly

3.5.4.3.2.1 Video processor assembly. This assembly shall contain the following:

a. Four sample and hold circuits for the peak detectors.
b. Arithmetic and normalization circuitry to convert four channel data into azimuth and elevation target position signals.
c. AGC control loop integrator circuits.
d. AGC discrete switching circuits for the preamplifier gain and pulse amplifier attenuator selection.
e. BIT GO/NO GO comparators.

3.5.4.3.2.2 Timing logic assembly. This assembly shall contain the following:

a. Master timing clock circuit.
b. Timing circuits for receiver control.
c. Timing circuits for correlator control.
d. Raw alarm synchronization circuit.

3.5.4.3.2.3 BIN logic assembly. This assembly shall contain:

a. Correlation logic and control circuits.
b. Loss of track logic.
c. Prefire pulse blanking circuit (not utilized).

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3.5.4.3.2.4 Track logic assembly. This assembly shall contain:

a. VPRF data storage, parity check and code fault generation circuits.
b. VPRF counter circuits for timing control at the VPRF selected.
c. Timing circuits for receiver control.
d. Timing circuits for Video Processor control.

3.5.4.4 Form factor. The overall dimensions, locations of connections, air inlets/outlets, ETI, etc., of the FAC Subsystem shall be in accordance with DL 1372AS 100.

3.5.4.5 Weight. The weight of the FAC RCVR shall not be greater than 5 pounds (2.3 kilograms) and the weight of the LRE shall not be greater than 10 pounds (5.6 kilograms).

3.5.4.6 Mounting. The FAC RCVR shall be mounted within the TSP on the FLIR RCVR optical housing. The mounting provisions shall permit the interchangeability of units meeting the alignment accuracy requirements without requiring the use of any boresighting equipment. The LRE shall be mounted in accordance with the requirements of DL 1372AS 100.

3.5.4.7 Electrical connections. Electrical connections shall be provided as necessary to interconnect this system to other related and associated systems conform to 3.3.3.1, and be in accordance with DL 1372AS 100.

3.5.5 Power supply (LVPS). The LVPS shall meet the following requirement.

3.5.5.1 Function. The LVPS shall convert 115/200 V 400 Hz, three-phase aircraft power to provide the necessary power forms to meet the requirements of all elements of the DRS.

3.5.5.2 Operating mode.

3.5.5.2.1 Turn-on. One signal voltage (LVPS ON) shall turn-on all LVPS outputs, except the 25 Vrms 360 Hz (Scan Drive) which shall require a separate command.

3.5.5.2.2 Turn-off. Removal of the signal voltage of Paragraph 3.5.5.2.1 shall turn-off all power outputs. Additionally, the LVPS shall turn-off all power outputs to the DRS whenever any of the following occurs:

a. Undervoltage in aircraft power input to the LVPS.
b. Over or undervoltage or overcurrent in regulated LVPS outputs.
c. Overcurrent in the unregulated LVPS outputs.

3.5.5.3 Contents. The LVPS shall contain the following:

a. +5.3 V, -200 V Linear Regulators
b. +6 V, -6.3 V Linear Regulators
c. +15 VDC Linear Regulators
d. ±25 VDC Linear Regulators
e. 25 Vrms 3 Phase AC Regulator
f. Input, Output
g. Rectifiers and Logic

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3.5.5.3.1 +5.3 V, -200 linear regulators. This subassembly shall provide +5.3 VDC and -200 VDC regulated power for the DRS. It shall also provide output over and undervoltage and overcurrent fault logic to the Rectifiers and Logic subassembly.

3.5.5.3.2 +6 V, -6.3 linear regulators. This subassembly shall provide +6 VDC and -6.3 VDC regulated power for the DRS. It shall also provide output over and undervoltage and overcurrent fault logic to the Rectifiers and Logic subassembly.

3.5.5.3.3 15 VDC linear regulators. This subassembly shall provide +15 VDC and -15 VDC regulated power for the DRS. It shall also provide output over and undervoltage and overcurrent fault logic to the Rectifiers and Logic subassembly.

3.5.5.3.4 25 VDC linear regulators. This subassembly shall provide +25 VDC and -25 VDC regulated power for the DRS. It shall also provide output over and undervoltage and overcurrent fault logic to the Rectifiers and Logic subassembly.

3.5.5.3.5 25 Vrms 3 phase AC regulator. This subassembly in conjunction with a power factor correcting network located in the Input, Output subassembly, shall provide 25 Vrms, 360 Hz, 3 Phase power for the Scan Mirror Assembly. It shall also provide output undervoltage and overcurrent fault logic to the Rectifiers and Logic subassembly.

3.5.5.3.6 Input, output. This subassembly shall provide external connectors, internal connectors and wiring, EMI input and output filtering and unit ON-OFF switching.

3.5.5.3.7 Rectifiers and logic. This subassembly shall provide as output voltages, inputs to each of the regulator subassemblies and overcurrent fault protected +28 VDC. This subassembly shall also provide a "shut-down" logic circuit responsive to the fault inputs from each of the regulator subassemblies. In addition, this subassembly shall provide 115 V 400 Hz 3 Phase aircraft input power undervoltage detection and ON-OFF logic.

3.5.5.4 Form factor. The overall dimensions of the LVPS including connector arrangement shall be in accordance with DL 1372AS 100.

3.5.5.5 Weight. The weight of the LVPS shall not exceed 37 pounds (17 kilograms).

3.5.5.6 Mounting. The LVPS shall be mounted in the aircraft in accordance with DL 1372AS 100.

3.5.5.7 Electrical connections. Electrical connections shall be provided as necessary to interconnect this system to the other related and associated systems conform to 3.3.3.1 and be in accordance with DL 1372AS 100.

3.5.6 Control, detecting-ranging set. The Control shall meet the following requirements:

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3.5.6.1 Function. The Control shall provide for the following:

a. DRS Power and Operating Modes
b. DRS Interlocks
c. FLIR Indicator Controls
d. DRS Over-temperature Junction
e. DRS BIT Function Control

3.5.6.2 Operating modes. The Control shall have the following operating modes.

3.5.6.2.1 DRS power and operating modes. Controls shall be provided for the DRS power and operating modes as specified in 3.5.6.3.3.1. Power and operational functions shall be provided as follows:

a. Turret and FLIR Power
b. Turret Modes
c. Laser Power
d. Laser Transmit Modes

3.5.6.2.1.1 Turret and FLIR power. The Turret and FLIR Power Switch (TUR/FLIR) shall provide three power functions, in each of the three switch postions labeled ON, STOW and OFF, as follows:

a. ON position - In this position, the TUR/FLIR Power Switch shall enable full power to the Stabilization Subsystem, FLIR Subsystem and Temperature Control. In addition, the Turret MODE Switch shall be activated.
b. STOW position - In this position, the TUR/FLIR Power Switch shall enable the TSP to rotate to the STOW position, then disable motors of the Stabilization Subsystem, except gyro spin motor. Power shall be applied to the Temperature Control and FLIR Subsystem except for the FLIR scanner and the Indicator video circuits.
c. OFF position - In this position, the TUR/FLIR Power Switch shall enable the TSP to rotate to the STOW position, then shall remove all power from Stabilization Subsystem, FLIR Subsystem and Temperature Control.

3.5.6.2.1.2 Turret modes. The Mode Switch shall provide three mode functions, in each of the three switch positions labeled COMPT (computer), MAN (manual), BRST (Boresight), as follows:

a. COMPT Position - In this position, the Mode Switch shall rotate the TSP as commanded by the Aircraft Computer.
b. MAN Position - In this position, the Mode Switch shall enable slewing of the TSP in response to the Bombardier/Navigator (B/N) Slew Stick commands to the Aircraft Computer.
c. BRST Position - In this position, the Mode Switch shall enable rotation of the TSP to the Boresight position (specified in 3.5.1.2).

3.5.6.2.1.3 Laser power. The Laser Power Switch (PWR) shall provide three power functions, in each of the three switch positions labeled ON, REC and OFF, when the TUR/FLIR switch is in the ON or STOW position. In addition, this switch

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shall be held in the ON position magnetically and shall return to REC position whenever power is removed from the magnetic hold in response to interlock functions specified in 3.5.6.2.2. The three switch positions shall function as follows:

a. ON Position - The Laser Subsystem shall be capable of firing and the FAC Subsystem shall be fully operational.
b. REC Position - The Laser Subsystem shall be powered but not capable of firing (see 3.5.3.2.1). The FAC Subsystem shall be fully operational.
c. OFF Position - The Laser Subsystem and the FAC Subsystem shall not be powered.

3.5.6.2.1.4 Laser transmit modes. With the Laser PWR switch in the ON position the Laser Subsystem shall be fully operational and, depending upon the position of the XMIT mode switch, shall operate in one of the following modes:

a. DES Position - In this position, the XMIT Switch shall enable Laser firing at a pre-selected variable pulse repetition frequency rate (VPRF) (see 3.5.3.2.3).
b. AUTO Position - In this position, the XMIT Switch shall enable an automatic selection of either Designate or Range mode functions as determined by the Aircraft Computer (see 3.5.3.2.4).
c. RANGE Position - In this position, the XMIT Switch shall enable a ranging function with VPRF determined by the Aircraft Computer (see 3.5.3.2.2).

3.5.6.2.2 Interlocks. Interlocks prohibiting DRS operation shall be provided as follows in accordance with the requirements of DL 1372AS 100:

a. (U) Receiver Group Service Door Interlock shall prevent Laser firing and Turret operation when open.
b. Landing Gear Interlocks (Nose Gear and Right Main Gear) shall prevent Laser firing when the landing gear is down.
c. Pilot Interrupt Interlock shall prevent Laser firing when depressed.
d. B/N Laser Inhibit switch (one switch) shall prevent Laser firing when activated.

3.5.6.2.3 Indicator controls. Controls shall be provided for following Indicator displayed signals:

a. Reticle Brightness
b. FAC Symbol Brightness
c. TV Contrast
d. Indicator Brightness
e. FLIR Gain
f. FLIR Focus
g. FLIR Polarity

3.5.6.2.4 Overtemperature junction. Overtemperature indication inputs from the WRAs listed below shall be routed to the Aircraft Temperature Control Relay as a single output. The DRS Control shall accept overtemperature indication from the following WRAs:

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a. LVPS
b. LTE/LRT
c. Compressor
d. ECA
e. Indicator

3.5.6.2.5 DRS BIT function control. The Control shall generate BIT command output signals to the DRS WRAs in response to input signals from the Aircraft BIT Panel, except when disabled in ATTACK mode.

3.5.6.3 Contents. The Control shall contain the following:

a. Decoder Module
b. Switching Module
c. Chassis Subassembly

3.5.6.3.1 Decoder module. This module shall provide for DRS BIT command logic in response to the Aircraft BIT Panel input selection. The subassembly shall also provide BIT GO/NO GO drive signals to the Aircraft BIT Panel.

3.5.6.3.2 Switching module. This module accepts major subsystem BIT selection (FLIR, Laser, Turret, DRS) from the Aircraft BIT Panel and shall provide power and logic functions required to support each test. The subassembly shall also provide a current limited "protected" 28 VDC source for Interlock Switching functions, the Aircraft BIT Panel selection voltage and for certain ECA functions.

3.5.6.3.3 Chassis subassembly. The Chassis shall provide a heatsink within the Aircraft Cockpit environment and shall provide for controls mounting and controls identification illumination.

3.5.6.3.3.1 Controls. Controls shall be provided as follows:

a. Power Controls
   1. Three position, pull-to-change, Turret and FLIR (TUR/FLIR) switch labeled ON, STOW and OFF.
   2. Three position Laser, pull-to-change, (LASER) switch labeled ON, REC and OFF. This switch shall incorporate magnetic latching and a mechanical guard.
b. Mode Controls
   1. Three position Turret, pull-to-change, (MODE) switch labeled COMPT, MAN and BRST.
   2. Three position Laser Transmit, pull-to-change, (XMIT) mode switch labeled DRS, AUTO and RANGE.
c. Indicator Controls (controls 1 through 6 shall be shape coded)
   1. Reticle Brightness potentiometer labeled RETICLE.
   2. FAC Symbol Brightness potentiometer labeled SYMBOL.
   3. TV Contrast potentiometer labeled TV CONTRAST.
   4. Indicator Brightness potentiometer labeled BRT.
   5. FLIR Gain potentiometer labeled GAIN.
   6. FLIR Focus potentiometer labeled FOCUS with "normal" detent.
   7. FLIR Polarity switch labeled BLACK and WHITE.

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3.5.6.4 Form factor. The overall dimensions of the Control, including final form factor, panel arrangement and connectors shall be in accordance with DL 1372AS 100.

3.5.6.5 Weight. The weight of the Control shall not exceed 5 pounds (2.3 kilograms).

3.5.6.6 Mounting. The Control shall be mounted in accordance with DL 1372AS 100.

3.5.6.7 Identification illumination. The Control lighting shall conform to the brightness and dimming requirements of 3.3.4.1 herein and DL 1372AS 100.

3.5.6.8 Electrical connections. Electrical connections shall be provided as necessary to interconnect Detecting-Ranging Control system to other related and associated systems as specified in DL 1372AS 100.

3.5.7 Control, temperature. The Temperature Control shall meet the following requirements:

3.5.7.1 Function. The Temperature Control shall sense the FLIR lens temperature and control the power to the TSP Heater Blanket which conditions the TSP air temperature. It shall maintain the FLIR lens temperature within the operational limits of the lens thermal compensation assembly for optimum focus.

3.5.7.2 Operating modes. The Temperature Control controls power to the Heater Blanket assembly by changing the duty cycle of the 115 V 400 Hz 3 Phase aircraft power. When FLIR lens temperature is above 75°F (24°C), the duty cycle is 0% (power off). For FLIR lens temperatures between 75°F and 70°F (21°C) the duty cycle varies linearly from 0% to 100%, with 50% duty cycle at 72.5°F (23°C) (nominal). For FLIR lens temperatures below 70°F a 100% duty cycle (full power) is generated, except, at initial turn on when a lower duty cycle is applied for 10 minutes (nominal) to prevent thermal shock to the Heater Blanket. All power shall be automatically inhibited in case of Heater Blanket overtemperature condition or element malfunction creating phase unbalance.

3.5.7.3 Contents. The Temperature Control shall contain the following:

a. Temperature Controller Subassembly
b. Temperature Controller Logic
c. Voltage Regulator +15 V

3.5.7.3.1 Temperature controller subassembly. This subassembly shall provide the main power relay, phase unbalance sense transformer, EMI filters and three (3) solid state switches for controlling the duty cycle. Also, it shall provide for externally mounted controls (see paragraph 3.5.7.3.4) and necessary wiring and connectors for Temperature Controller Logic and Voltage Regulator.

3.5.7.3.2 Temperature controller logic. This subassembly shall provide FLIR lens temperature sense signal processing and it shall generate temperature proportional control drive for the three solid state switches in the Temperature Controller Subassembly. Also, it shall contain circuitry for automatic power shut off to the Heater Blanket due to overtemperature or element malfunction. BIT voltage source shall be located on this subassembly.

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3.5.7.3.3 Voltage regulator +15 V. This subassembly shall provide regulated +15 VDC and rectifier load circuit used in detection of phase unbalance.

3.5.7.3.4 Controls. The Temperature Control shall provide for externally mounted controls as follows:

a. OVERTEMP RESET Switch to reset the BIT function capability of the Temperature Control.
b. BIT Switch to activate BIT circuitry and indicate a GO condition by illuminating the BIT GO Indicator on the Temperature Control.

3.5.7.4 Form factor. The overall dimensions of the Temperature Control, including final form factor, external controls arrangement and connectors shall be in accordance with DL 1372AS 100.

3.5.7.5 Weight. The weight of the Temperature Control shall not be greater than 7 pounds (3.2 kilograms).

3.5.7.6 Mounting. The Temperature Control shall be mounted in the aircraft in accordance with DL 1372AS 100.

3.5.7.7 Electrical connections. Electrical connections shall be provided as necessary to interconnect this system to other related and associated systems, conform to 3.3.3.1, and be in accordance with DL 1372AS 100.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the procuring activity. The procuring activity reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Finished article tests, qualification tests, and reliability and maintainability demonstrations shall be performed only if specifically or separately provisioned.

4.1.1 Classification of tests. Items covered by this specification shall be capable of passing the tests specified herein to determine compliance with all applicable requirements. These tests shall consist of the following:

a. Manufacturing Reliability Screening (see 4.2)
b. Acceptance Tests (see 4.3)

4.2 Manufacturing reliability screening. Each circuit board (card) and Items a. through n. of 1.2 shall be subjected by the contractor to manufacturing reliability screening as described herein. The contractor shall advise the Navy on-site representative as to when screenings are to commence so that a representative may be designated to witness the screens on a not-to-interfere basis, when so desired.

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4.2.1 Card conditioning (non-operating) thermal. Each card shall be subjected to a minimum of 10 temperature cyclings of -40 to +110 to -40°C at a rate of change of not less than 10°C per minute (the temperature range may be adjusted by the contractor provided the range from minimum to maximum temperature is not less than 90°C). The contractor shall analyze any failure trends that occur and if appropriate, define, implement, and verify any necessary correction action.

4.2.2 Unit vibration. Each unit (Items a. through n. of 1.2) shall be subjected to complex/random vibration as specified in its screening ARD. Each unit shall have power on during the screening, and there shall be continuous monitoring of the unit operation. The contractor shall analyze any failures that occur and define, implement and verify any necessary correction action.

4.2.3 Unit thermal. Each unit (Items a. through n. of 1.2 except the compressor, Item e.) shall be subjected to a temperature cycles as specified in its screening ARD. Each unit shall have power on during periods of increasing temperature and power off during periods of decreasing temperature. During the power on periods, there shall be continuous monitoring of the unit operation. Where applicable, during the screening the covers shall be removed from the unit, there shall be no cooling air other than chamber air, and a circulating fan shall be in the chamber. The contractor shall analyze any failure trends that occur and if appropriate, define, implement, and verify any necessary correction action.

4.3 Acceptance tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government representative. The Navy on-site representative shall be advised when system level failure free MRI tests are to commence so that a representative may be designated to witness the tests, when so desired. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall maintain test data that shall include a tabulation of all failures which occurred during these tests and for system level tests an indication of whether or not the BIT of the equipment confirmed and/or isolated the failure. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual Tests
b. Reliability Assurance Tests

4.3.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each equipment accepted shall have passed the following tests:

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a. Examination of Product.
b. Operational Tests (Unit and System).
c. Manufacturing Run-In Test (MRI).

4.3.1.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met and that all internal components and subassemblies are properly coded, located and affixed, and that all external devices such as flanges, mounting provisions, and connector locations are correct as specified herein.

4.3.1.2 Operational tests (unit and system). Unit and system operational tests shall be performed in accordance with equipment acceptance requirement documents (ARDs) in effect at the time of test and data shall be obtained to establish full compliance with unit and system ARD performance requirements.

4.3.1.2.1 Unit operational tests. Prior to assembly of a DRS each unit (Items a. through n. of 1.2 herein) shall be subjected to operational tests. These tests shall be devised to verify that each unit will function satisfactorily (without adjustments of controls not accessible to the operator in normal flight operation) when assembled as a operational DRS. For each unit the signals and functions necessary to assure adequate system operation shall be measured.

4.3.1.2.2 System operational tests. System tests shall be devised to verify that the total system functions properly in a fully operational configuration to the accuracies specified in equipment acceptance requirement documents.

4.3.1.3 Manufacturing run-in (MRI) test. Each unit (Items a. through n. of 1.2 herein) shall be subjected by the contractor to the following MRI tests. These tests shall follow completion of the manufacturing reliability screening of 4.2. The contractor shall advise the procuring activity as to when tests are to commence so that a representative may be designated to witness the tests, when so desired.

4.3.1.3.1 Unit vibration. Each unit (Items a. through n. of 1.2 herein) shall successfully complete 10 minutes of failure-free operation as specified in its screening ARD. Each unit shall have power on during the test, and there shall be continuous monitoring of the unit operation. Should a failure occur, it shall be repaired and the test started over at the beginning of the failure-free test period until the failure-free requirement is met.

4.3.1.3.2 Unit thermal. Each unit (Items b. through n. of 1.2 herein) shall successfully complete failure-free operation as specified in its screening ARD. Each shall have power on during periods of increasing temperature and power off during periods of decreasing temperature. During the power on periods, there shall be continuous monitoring of the unit operation. Where applicable during the test, the covers shall be removed from the unit, there shall be no cooling air other than chamber air, and a circulating fan shall be in the chamber. Should a failure occur, it shall be repaired and the test started over at the beginning of the first cycle until the three cycle failure-free requirement is met.

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4.3.2 Reliability assurance tests. Reliability assurance tests shall be conducted in accordance with 4.3.2.1 and 4.3.2.1.1 herein. Classification of failures shall be in accordance with 4.3.4.

4.3.2.1 Reliability production screening tests. Each system, throughout production, shall be tested to test level E for cockpit mounted units and test level F for all others as contained in MIL-STD-781 and as modified in 4.3.2.1.1 below.

4.3.2.1.1 All equipment screening test. Each system shall successfully complete 8 hours/two cycles of failure-free operation. Prior to this each system shall be subjected to not less than 8 hours/two cycles of burn-in. Should the test be stopped for any DRS failure during the failure-free portion of the cycle, the test shall be started over at the beginning of the first cycle until the failure-free requirement is met. In the event that a system experiences seven restarts, for any DRS failure, during the failure-free operation, the procuring activity on-site technical representative shall be notified immediately.

4.3.2.2 Test details. The test details, such as the length of the cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the screening ARDs.

4.3.2.3 Determination of reliability assurance compliance. Compliance criteria shall be as specified in the applicable ARD and determination of reliability assurance compliance shall be demonstrated during the all equipment screening test described in 4.3.2.1.1.

4.3.2.4 Duty cycle. For purposes of reliability (MTBF) testing a 10 percent duty cycle shall be applicable to the LRT and a 100 percent duty cycle applicable to all other components.

4.3.3 Failure corrective action. Should a failure occur during any test, the following action shall be taken:

a. Determine the cause of failure.
b. Classify the failure in accordance with MIL-STD-2074.
c. Determine if the BIT functioned properly to detect and isolate the failure.
d. Take appropriate corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

4.3.4 Failure definition and classification. The following failure definition and classification shall apply during the reliability assurance tests, 4.3.2.

4.3.4.1 Failure. Whenever any of the performance characteristics are outside of the requirements of the specification at any specified environmental conditions, at least one failure has occurred.

4.3.4.2 Failure classification. All failures occurring during formal reliability assurance testing shall be classified as either relevant or nonrelevant. Only those failures classified relevant shall be used for making an accept/reject decision.

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4.3.4.2.1 Relevant failure. A failure occurring during formal reliability assurance tests, which is not defined in 4.3.4.2.2, shall be considered relevant with the following exceptions:

a. Multiple failures - In the event simultaneous part failures occur, each failed part which would independently prevent satisfactory equipment performance shall be counted as a relevant failure except if the contractor and the procuring activity agree that the failure of one part was entirely responsible for the failure of any other parts, then each such dependent part failure shall not be counted as a relevant failure.
b. Intermittent failures - The second (and any subsequent) occurrences of the same intermittent failure on the same unit shall be counted as one relevant failure.
c. Failures of built-in-test - Any malfunction (including a false alarm) of the BIT features of the equipment shall be classified as a relevant failure.

4.3.4.2.2 Nonrelevant failures. Only those failures listed below may be counted as nonrelevant:

a. Failures directly attributable to improper installation in the test chamber.
b. Failures of test instrumentation or monitoring equipment (other than the BIT function).
c. Failures resulting from test operator error in setting up or in testing the equipment.
d. A dependent failure resulting from a nonrelevant failure.
e. Failures attributable to an error in the test procedures.
f. Failures occurring during troubleshooting, repair, verification, or set-up time.
g. A failure of any of the following items:
    1. The LRT events counter.
    2. The LRT nitrogen pressure gage if the mode of failure does not affect DRS operation.
    3. Time totalizing meters.
    4. Compressor Pressure Gage
    5. A single lamp failure of redundant status lamp on the indicator illuminated panel and Temperature Control self test lamp.
h. Failures clearly attributable to an overstress condition in excess of the design requirements.
i. Adjustments of controls which are accessible to the operator during normal use shall be counted as nonrelevant failures provided the adjustment is accomplished without reference to test equipment, meters, indicators, etc., which are not a functional part of the equipment under test.
j. FLIR detector channel failures that occur subsequent to the established test baseline shall be classified in accordance with the requirements of the system specification utilizing the same FLIR brightness, gain and detail settings recorded during the establishment of the test baseline. In event of a disagreement regarding the baseline measurements, the Failure Review Board shall adjudicate the disagreement.

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k. For those failures where corrective action has been approved by the procuring activity prior to end of test but corrective hardware is not available for the system under test, subsequent failures of uncorrected portions shall be classified nonrelevant.
l. The first occurrence of an intermittent failure on any one DRS system. A one-time failure which cannot be verified or re-induced with a mutually agreeable amount of troubleshooting is defined as an intermittent failure.
m. The subsequent failures of a "pattern failure".
n. Failure or replacement of limited life items subsequent to stipulated life.
o. Failures not occurring in formal test. Formal test is recognized as the actual time during an officially designated cycle.

4.3.4.3 Retest. If the test results in a reject decision or if pattern failures are present, corrective action shall be accomplished in accordance with MIL-STD-781 and, using the same sample size, a retest shall be conducted at the direction of the procuring activity.

4.3.4.4 Failure analysis. An analysis of each failure shall be made.

4.4 Test procedures. MIL-T-18303 shall be used as a guide for preparation of test procedure specified in appropriate ARDs. When approved ARDs are available from previous contracts, such ARDs may be used when their use is approved by the designated government representative.

4.5 Reconditioning of equipment under test. DRSs which have successfully passed acceptance tests and require reconditioning shall have all worn and damaged items replaced by the contractor. After reworking, the contractor shall resubmit the equipment for acceptance. The flashlamp and laser rod shall be considered worn when 50 percent of their life has been exceeded. Equipment that has been subjected only to acceptance testing shall not require MRI retesting following replacement of worn limited life items.

4.6 Presubmission testing. No item, part, or DRS shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable lower tier requirements.

4.7 Rejection and retest. DRSs which have been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, particulars concerning previous rejection and the action taken to correct the defects shall be documented on the Quality History Record and made available to the procuring activity on request. Replacement components, SRAs, or WRAs may be substituted for failed components, SRAs, or WRAs at any level of Manufacturing Reliability Screening and Acceptance Testing provided that any replacement SRA has completed its SRA level tests and any replacement WRA has completed its WRA level tests.

5. PACKAGING

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for level A

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shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended use. The DRS, when installed in the A-6E TRAM aircraft, will provide an autonomous night and limited weather weapon delivery capability.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1).

6.3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence should apply:

a. Contract - The contract should have precedence over any specification.
b. This specification - This specification should have precedence overall applicable subsidiary specifications. Any deviation from this specification or from subsidiary specifications, where applicable, should be specifically approved in writing by the procuring activity.
c. Referenced specifications - Any referenced specifications should have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications should apply to the extent specified.

6.4 Performance objectives. Minimum size and weight, simplicity of operation, of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which should be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval should be submitted to the procuring activity for consideration. Each request should be accompanied by complete supporting information.

6.5 Non-repairable subassemblies. As a general rule, non-repairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts, tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. The non-repairable subassembly should evidence an MTBF greater than that of a similar repairable one.

6.6 Associated equipment. The DRS shall operate with the following associated equipment which should not be supplied as part of the equipment:

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a. Control Panel, Pilots Control Box (PCB)
b. Ballistics Computer Set, AN/ASQ-155 (Aircraft Computer)
c. A-D, D-A Converter, CV-3163/ASQ-155
d. Computer Control, C-9535/ASQ-155
e. Indicator Group Vertical Display Subsystem, AN/AVA-1
f. Analog Display Indicator, IP-722D/AVA-1 (ADI)
g. Video Tape Recorder (VTR), AN/USH-17
h. Search Radar, AN/APQ-156
i. Radar Set Control, C-9436/APQ-156 (BNCB)
j. Fault Locating Indicator, ID-1933/APQ-156 (BIT Panel)
k. WALLEYE Subsystem
l. CONDOR Subsystem
m. BIT Panel

6.7 Data requirements. No deliverable data is required by this specification.

6.8 List of acronyms or abbreviations.

AC - Alternating Current
A/D - Analog-to-Digital
ADI - Analog Display Indicator
ADL - Armament Data Line
AGC - Automatic Gain Control
ARC - Automatic Responsivity Control
ARD - Acceptance Requirement Document
Az - Azimuth
BIT - Built-in-Test
BITE - Built-in-Test Equipment
B/N - Bombardier/Navigator
BRST - Boresight
CRT - Cathode Ray Tube
CG - Center of Gravity
cm - Centimeter
COMP - Computer
CW - Continuous Wave
DC - Direct Current
DES - Designate
DRS - Detecting - Ranging Set, AN/AAS-33
ECA - Amplifier, Electronic Control
EL - Elevation
EMI - Electromagnetic Interference
ETI - Elapsed Time Indicator

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FAC	-	Forward Air Controller
FIT	-	Fault Isolation Test
FLIR	-	Forward Looking Infrared
FOV	-	Field of View
GAI	-	Gimbal Angle Indicators
GSE	-	Ground Support Equipment
Hz	-	Hertz
IFOV	-	Instantaneous Field of View
ICD	-	Interface Control Document
IG	-	Inner Gimbal
in	-	Inch
IR	-	Infrared
IMC	-	Image Motion Compensation
FAC RCVR	-	Receiver, Laser
LFOV	-	Linear Field of View
LOS	-	Line-Of-Sight
LRT	-	Receiver-Transmitter, Laser
LRE	-	Processor, Signal
LTE	-	Generator-Processor
LVPS	-	Power Supply, Low-Voltage
MAN	-	Manual
MDA	-	Motor Drive Amplifier
MIC	-	Microelectronic Integrated Circuit
MRI	-	Manufacturing Run-In
MRT	-	Minimum Resolvable Temperature
MTBF	-	Mean-Time-Between Failure
MTF	-	Modulation Transfer Function
MTTR	-	Mean-Time-To-Repair
NARC	-	Narcissus Assembly
NET	-	Noise Equivalent Temperature
NFOW	-	Narrow Field of View
ORL	-	Optical Reference Line
ORT	-	Operational Readiness Test
PCB	-	Pilots Control Box
PCU	-	Pedestal Control Unit
PFN	-	Pulse Forming Network

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PIM - Pulse Interval Modulation
pps - Pulses Per Second
PRF - Pulse Repetition Frequency
QPL - Qualified Parts List
REC - Receive
RDT - Reliability Demonstration Test
RG - Receiver Group
SDC - Signal Data Converter
SITS - System Integration Test Station
S/N - Signal-to-Noise Ratio
SRA - Shop Replaceable Assembly
TSP - Turret, Stabilized Platform
TTL - Transistor-to-Transistor Logic
TUR - Turret
VPRF - Variable Pulse Repetition Frequency
Vrms - Volts root-mean-square
VTR - Video Tape Recorder
WFOV - Wide Field of View
WRA - Weapon Replaceable Assembly

6.9 Definitions

6.9.1 Failure. Failure definition and classification of failures is as defined in 4.3.4 of this specification.

6.9.2 Interchangeability. Interchangeability is as defined in MIL-E-5400 except that no calibrations or adjustments shall be permitted on a system basis, but only on WRAs at intermediate level.

6.9.3 WRA. A WRA, as used herein, is an item which can be removed and replaced within the end item (aircraft) to repair a malfunction in the most expeditious and economical manner. As such, a WRA may be a part, component, subassembly or assembly, e.g., fuse, printed circuit card, receiver, transmitter, relief valve, actuator, etc.

6.9.4 Subassembly. Refers to the first level of mechanical disassembly within the WRA. A subassembly is two or more parts which form a portion of a higher assembly, but which is removable as a whole. It may have a part or parts which are individually replaceable.

6.9.5 Contractor. Where the word contractor is used herein, it should refer to the Government subcontractor who has the responsibility for the design, construction and test of the equipment covered by the applicable equipment design control specification.

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6.9.6 Optical reference line (ORL). The ORL is defined as the LOS indicated by the FLIR reticle in the narrow FOV.

6.9.7 Zero reference position. The zero reference position is defined as that set of gimbal angles which will align the optical reference line to the ADL of the aircraft.

6.9.8 (U) Platform. Any stabilizing system for the Optical Reference Line.

Preparing Activity:
Navy - AS
(Project Number 5855-N-165)

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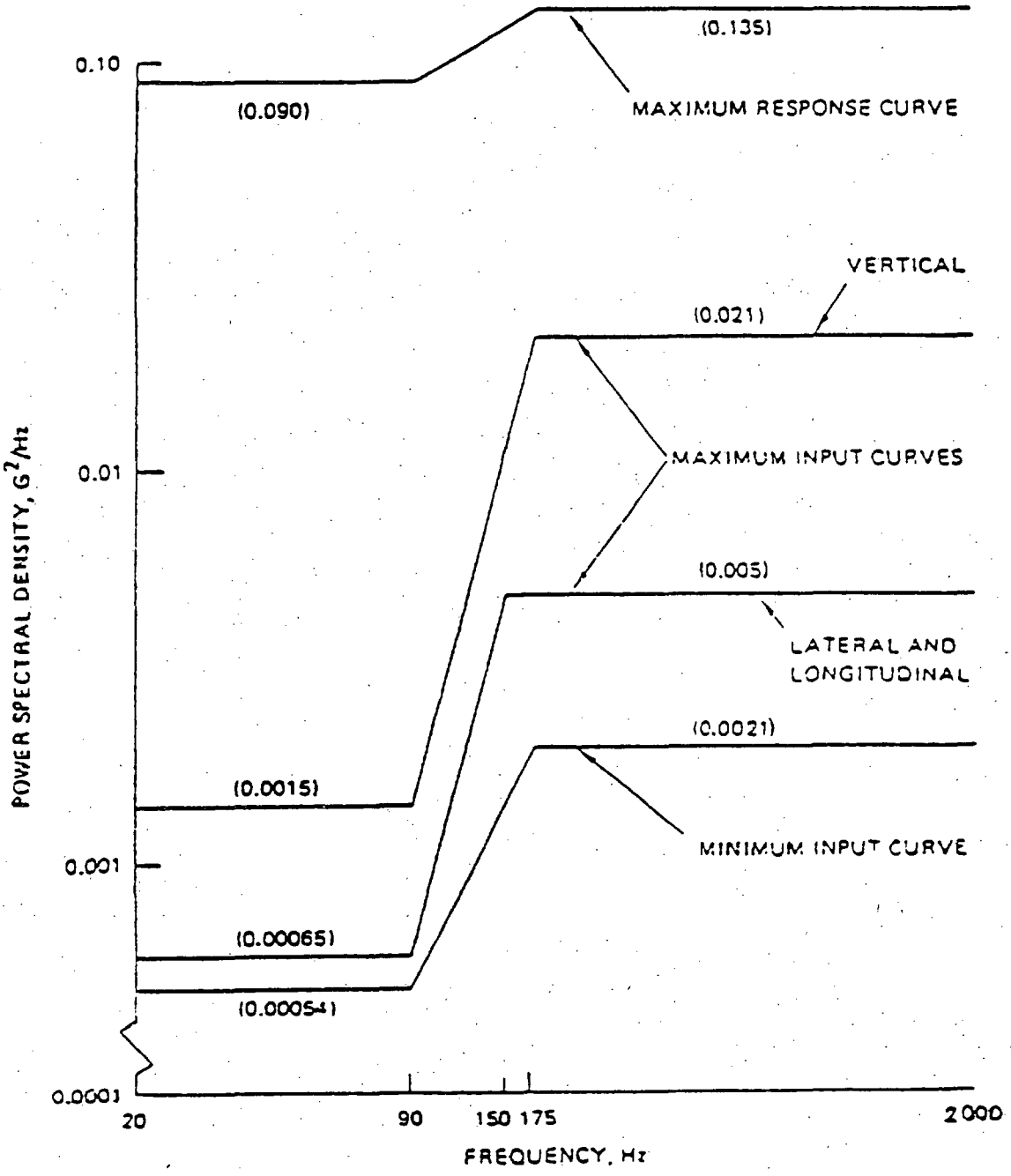


FIGURE 1. Random vibration spectrum, receiver group.

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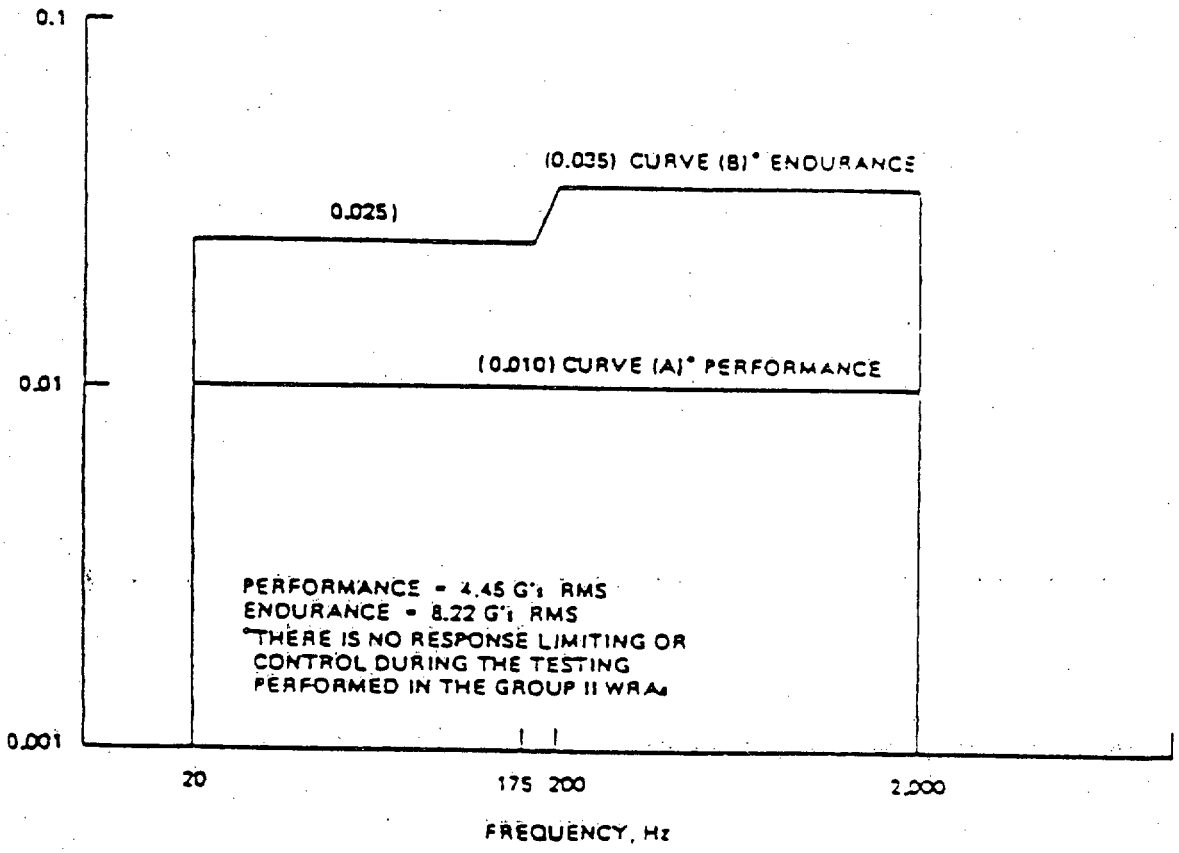


FIGURE 2. Random vibration spectrum, group II WRAs.

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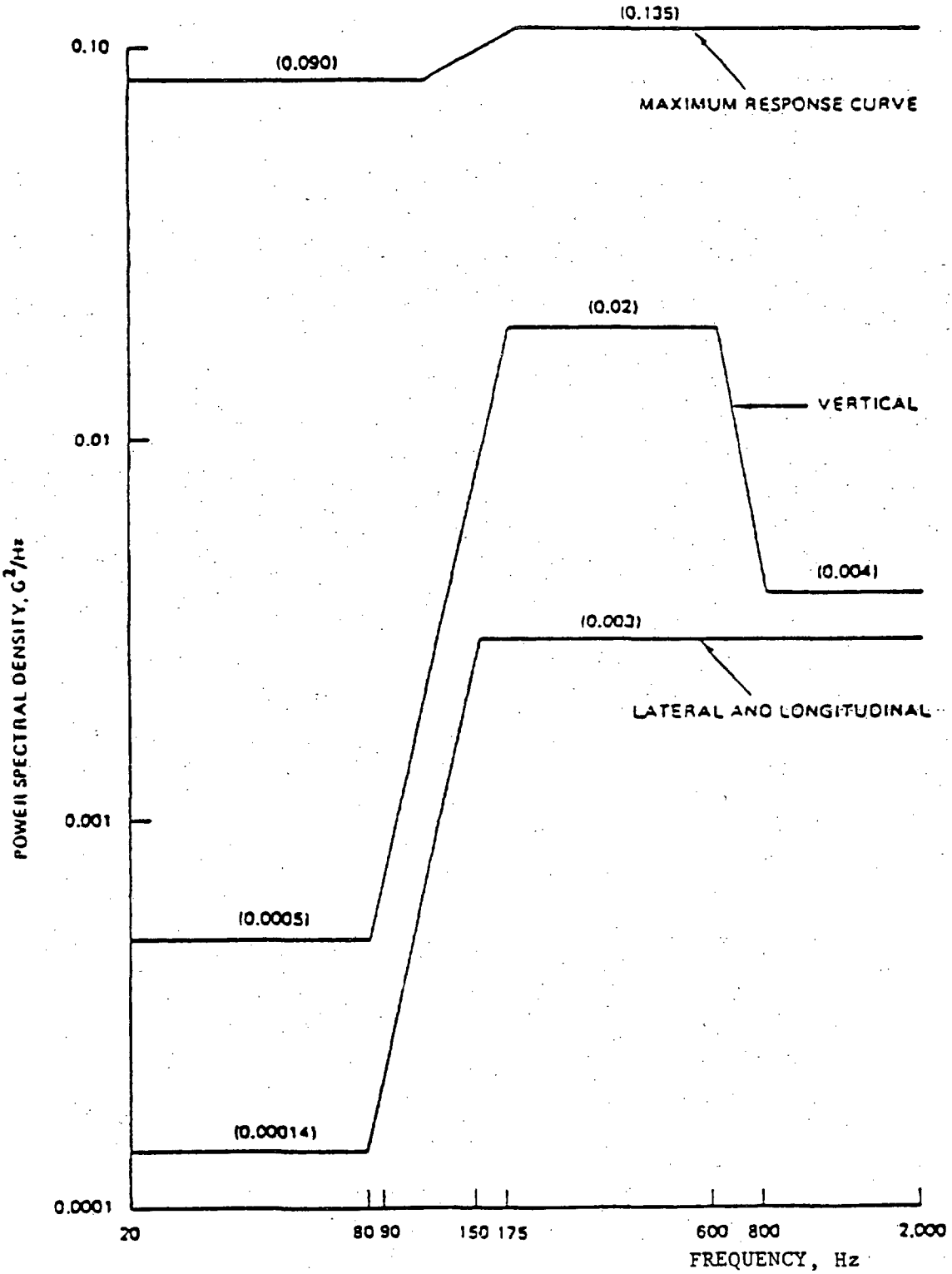


FIGURE 3. Flight vibration envelope, receiver group.

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Altitude	
Feet	Meters
SL	SL
10,000	3,048
20,000	6,096
30,000	9,144
40,000	12,192
50,000	15,240

Temperature	
^{o}F	^{o}C
-100	-73
-50	-46
0	-18
+50	+10
+100	+38
+150	+66
+200	+93

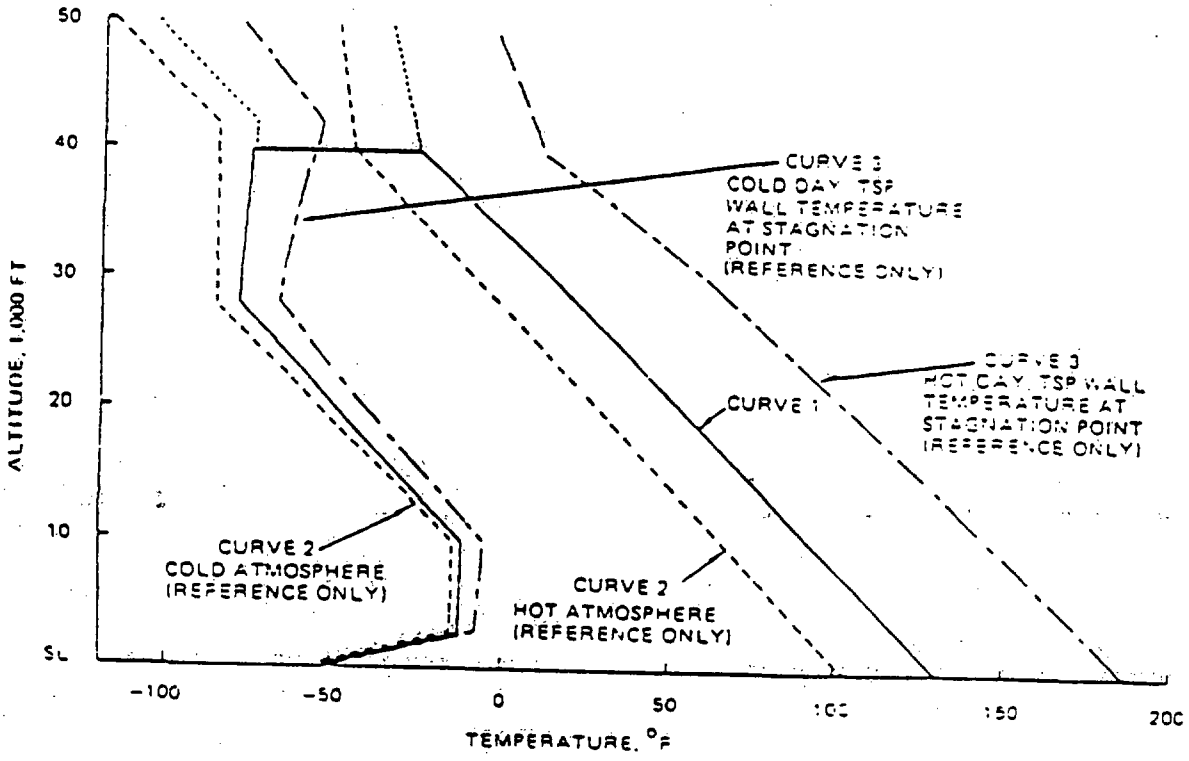


FIGURE 4. Receiver group operating envelope.

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Temperature	
°F	°C
-80	-62
-60	-51
-40	-40
-20	-28
0	-18
+20	-7
+40	+4
+60	+16
+80	+27
+100	+38
+120	+49
+140	+60
+160	+71
+180	+82

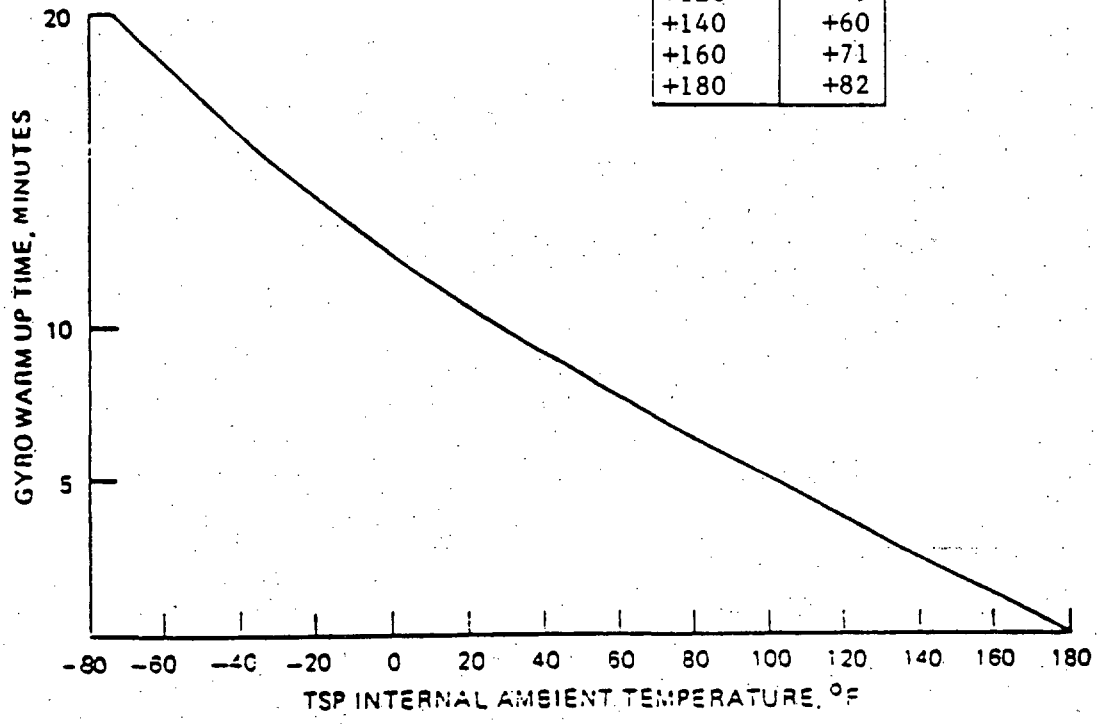
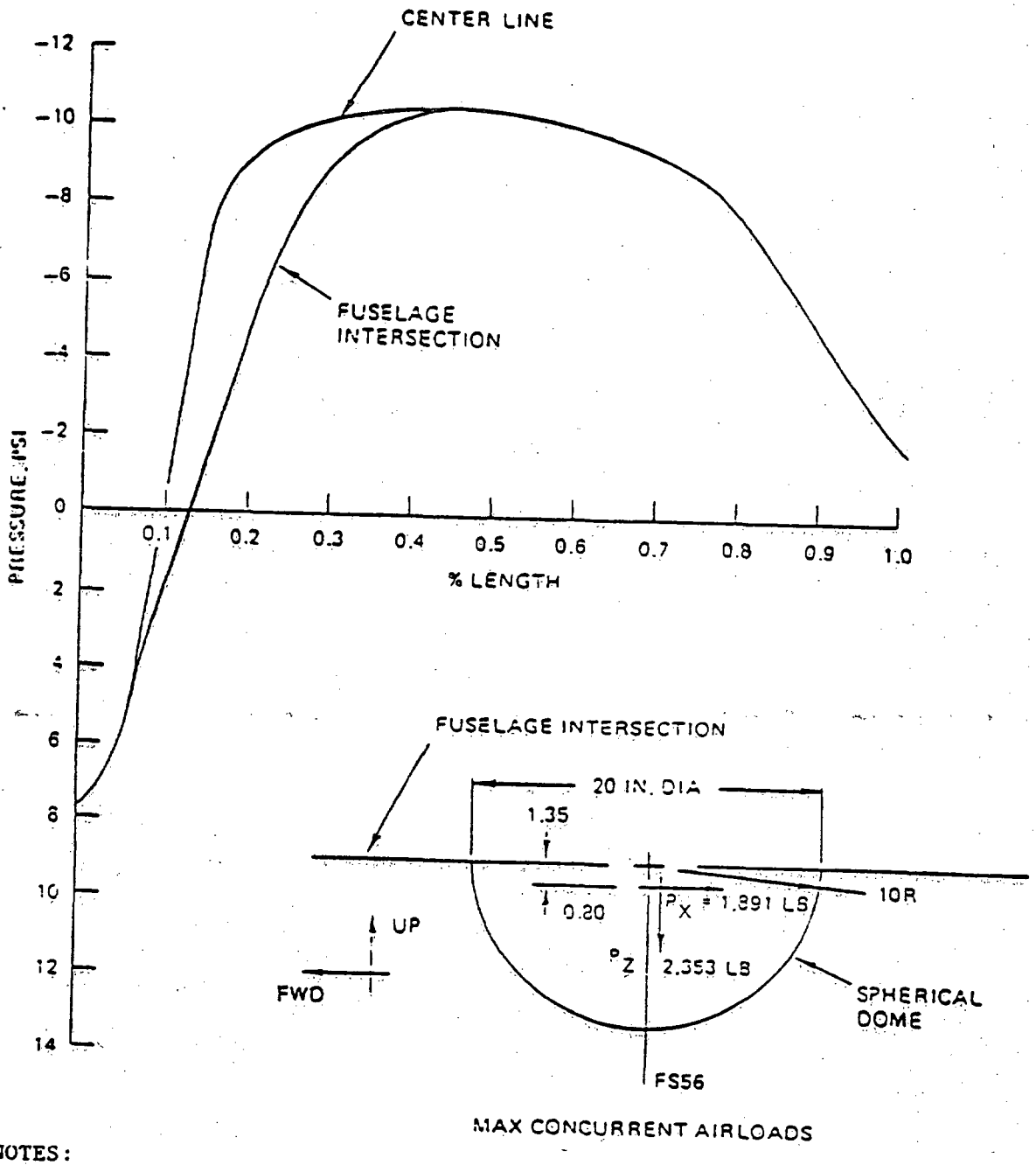


FIGURE 5. Gyro warm-up time.

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NOTES:

1/ FOR YAWED FLIGHT P_X IS PARALLEL TO RELATIVE WIND
MAX BETA FOR ABOVE LOADING IS 5.0 DEG

2/ SEE 3.3.14.2 FOR LIMIT LOAD FACTORS

FIGURE 6. Maximum airload and pressure distribution.

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Dimensions

in	cm
1/16	.16
1/8	.32
0.93	2.36
1.20	3.05
1.84	4.67
2.22	5.64
2.84	7.21
3.20	8.13
3.42	8.69

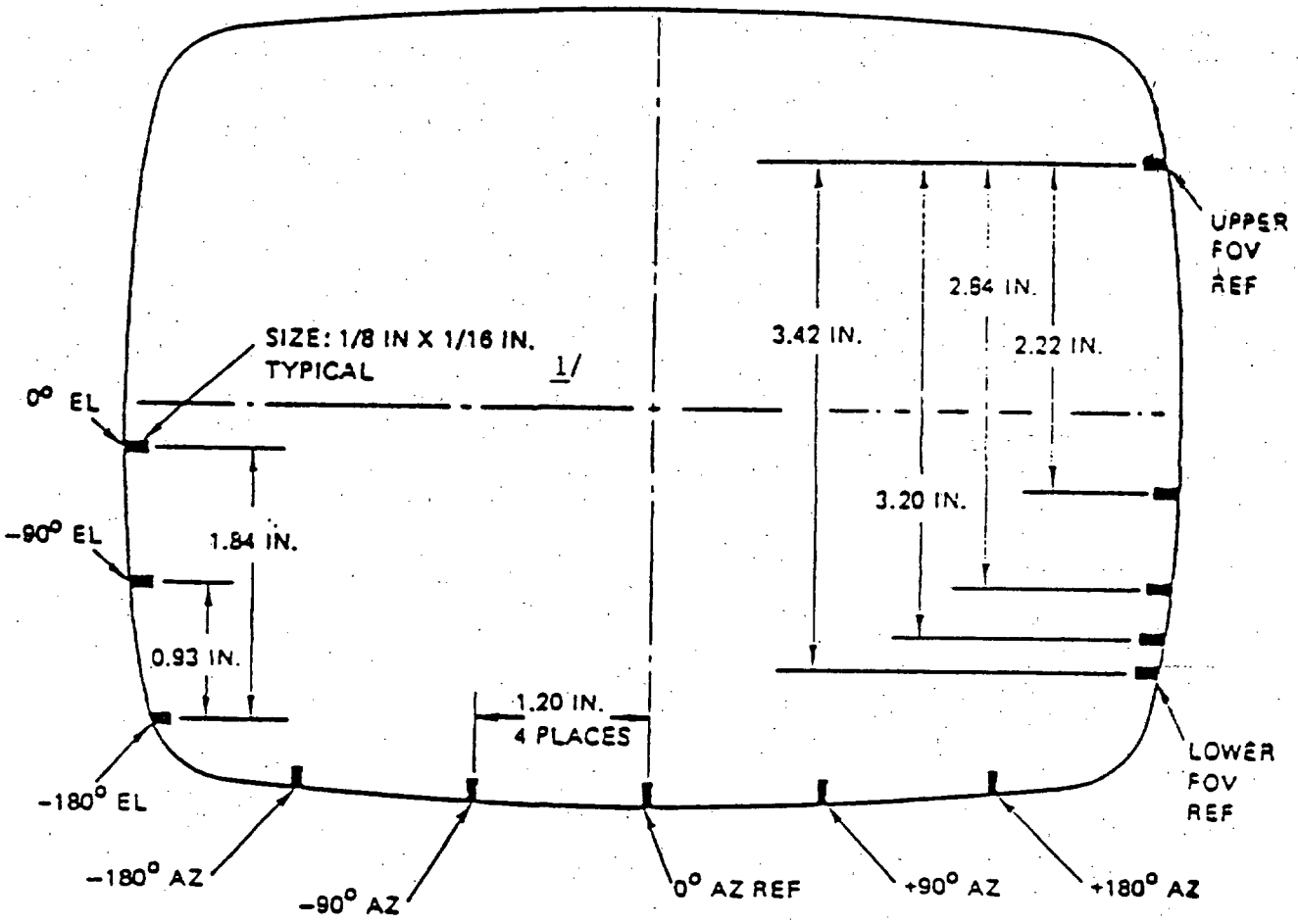
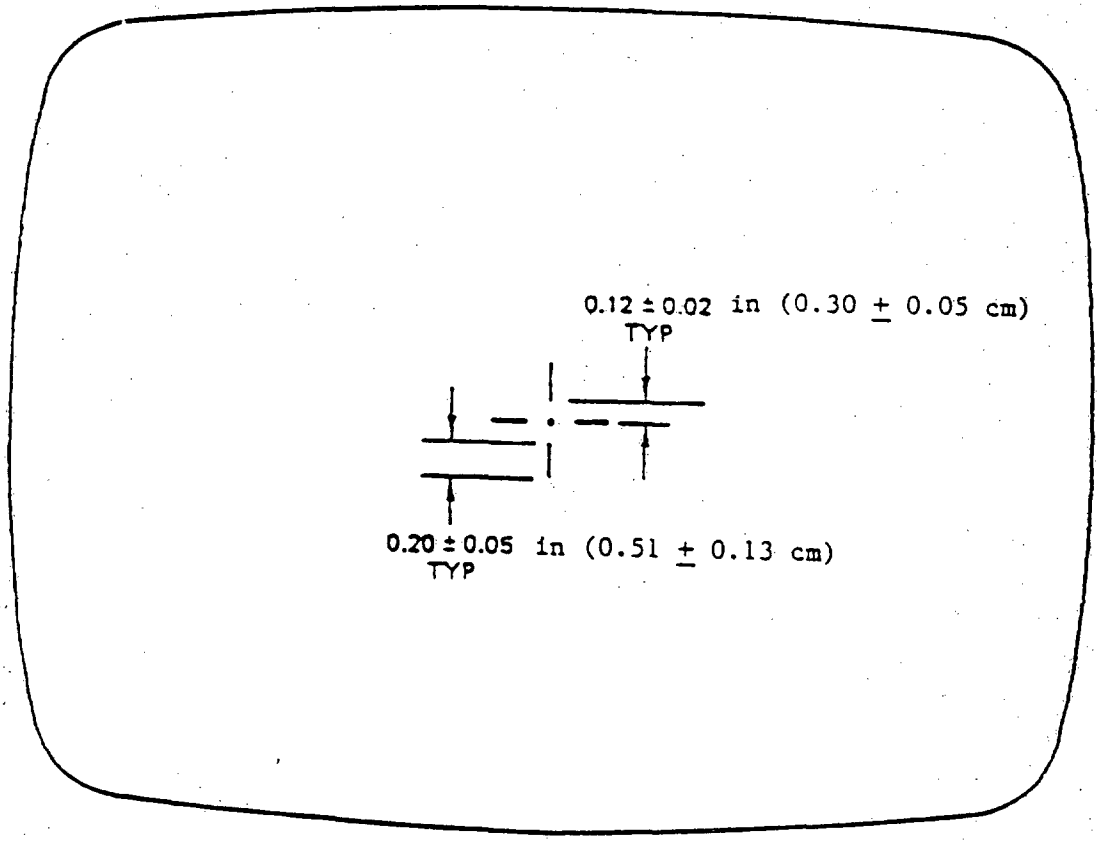


FIGURE 7. Reference markers.

NOTE:
1/ All dimensions approximate

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NOTE:

1/ THICKNESS OF ALL LINES AND CENTER DOT OF RETICLE SHALL BE EQUAL TO OR LESS THAN 3/64 INCH (0.12 cm)

FIGURE 8. Reticle characteristics.

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

Detecting-Ranging Set AN/AAS-33A MIL-D-85445(AS)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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